MILITARY SPECIFICATION

CENTRAL MIXING PLANT, CONCRETE, MOBILE, 250 CUBIC YARDS PER HOUR CAPACITY

1. SCOPE

1.1 This specification covers a mobile, self erecting, concrete central mixing plant for dry ready mix, and single or dual wet mixing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-S-300	- Sheeting and Tape, Reflective; Nonexposed Lens, Adhesive Backing.
FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
HH-L-361	- Linings; Brake (Automotive-Use).
QQ-S-781	- Steel, Strapping, Flat.
QQ-S-790	- Steel Strapping, Round (Bare and Zinc-Coated).
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

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Military

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-V-1174 - Varnish, Spar, Water Resisting (Formula No. 80).
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-3600 - Compressor, Rotary, Power-Driven; and Compressor, Recriprocating, Power Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-3859 - Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14-Ounce and 21-Ounce Capacity).
MIL-H-3992 - Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Aluminum and Aluminum Alloys.
MIL-S-8660 - Silicon Compound.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-R-11461 - Rod, Ground (with Attachments).
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.

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MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424	- Hand Tools; Packaging of.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-N-25027	- Nut, Self Locking, 250° F., 450° F., and 800° F., 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU.
MIL-L-45199	- Lubricating Oil, Internal Combustion Engine (High Output Diesel).
MIL-P-46093	- Primer Coating, Synthetic (for Brake Drums).
MIL-C-52058	- Chains, Roller, Power Transmission, and Conveyor.
MIL-C-52286	- Cable and Lead Assemblies, Power, Electrical, 300 to 600 Volts, 5 to 230 Amperes.
MIL-W-52574	- Welding and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-281	- Automobile, Trucks, Truck-Tractors, Trailers, and Trailer Dollies; Preservation and Packaging of.

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MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-621	- Subgrade, Subbase, and Test Method for Pavement Base-Course Materials.
MIL-STD-642	- Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1224	- Visual Inspection Guide for Pneumatic Tires (Nonaircraft).
MS19081	- Bearings, Roller, Tapered, Single Row of Rollers, Normal Angle.
MS21322	- Clamps, Loop - Aluminum - Plain and Cushioned.
MS21333	- Clamps, Loop - Steel - Plain and Cushioned.
MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS27142	- Connector, Plug, Electrical - Pin Contact, No. 12, 14, & 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 & 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way, No. 12, 14 & 16 A.W.G., Waterproof.
MS27148	- Contact; Pin, Electrical Connector: No. 12, 14, & 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35387	- Reflector, Indicating, Clearance.
MS35390	- Tire, Pneumatic: Low Platform Trailer, Earthmover, Grader, and Amphibious Vehicle.

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MS35392	- Inner Tube: Pneumatic Tire.
MS35423	- Light, Marker, Clearance - Service.
MS35424	- Light, Marker, Clearance - Blackout.
MS35647	- Padlocks, Key Operated.
MS35746	- Coupling, Automotive: Air Brake Lines.
MS35844	- Plug, Machine Thread, Magnetic: (Drain).
MS35865	- Axle, Vehicular, Non-Driving, Single, for Tactical Wheeled Vehicles.
MS35866	- Axle Assembly, Vehicular Non-Driving, Tandem for Tactical Wheeled Vehicles.
MS39020	- Band, Marker, Blank.
MS39062	- Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS39267	- Hose Assembly, Rubber: Hydraulic, Pressure Type, Minimum Bend Radius (for Engineer Equipment).
MS39317	- Hose Assemblies; Hydraulic, Double Wire Braid Reinforcement, with Reusable Couplings.
MS39318	- Hose Assemblies; Hydraulic, 4-Spiral Wrap Reinforcement, with Reusable Couplings.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51330	- Stop Light- Taillight, Vehicular- 24 Volt, Blackout Tail, Blackout Stop.
MS51331	- Guard, Splash, Wheel.
MS51337	- Coupler, Drawbar, Ring - Heavy Duty.
MS53004	- Valve, Relay, Emergency-Breakaway, Air Brake.
MS53007	- Plate, Identification - Emergency and Service.
MS53036	- Kingpin, Fifth Wheel, 2 Inch.
MS75020	- Connector, Plug, Electrical - 12 Contact Intervehicular, 24 Volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS500008	- Truck Tractor, 5 Ton, 6 X 6, M52.

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A48	- Gray Iron Castings.
B117	- Method of Salt Spary (Fog) Testing.
F52	- Methods of Test for Silting Index of Fluids for Processing Electron and Microelectronic Devices.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules.
E-3 - Electrical Lighting and Wiring for Commercial Vehicle Bodies.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

INTERSTATE COMMERCE COMMISSION

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

TIRE AND RIM ASSOCIATION, INCORPORATED

Tire and Rim Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Command Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

Standard MG1 Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017.)

CONCRETE PLANT MANUFACTURERS BUREAU

CPMB Standards

(Applications for copies should be addressed to the Concrete Plant Manufacturers Bureau, 900 Spring Street, Silver Spring, Maryland 20910.)

NATIONAL BOARD OF FIRE UNDERWRITERS

NFPAN070 - National Electrical Code.

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York, N. Y. 10038.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

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UNITED STATES OF AMERICA STANDARDS INSTITUTE

B6.6 - Gear Tolerances and Inspection.
B55.1 - Specification for Multiple V-Belt Drives.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10017.)

STATE OF CALIFORNIA

Vehicle Code of California.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, California 95809.)

FORGING INDUSTRY ASSOCIATION

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, Cleveland, Ohio 44101.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC.

RP 3-1-12 - Trailer, Air-Brake System Test.
RP 6-2-9 - Mounting of Required ICC Lighting Devices.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1413 K Street NW, Washington, D. C. 20005.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The mobile concrete central mixing plant, hereinafter called the "plant," consists essentially of a mixer base section with provisions for positioning two mixer sections in the operating position, an aggregate section, and a cement section, together with component systems and accessories. All sections of the plant are semitrailer or trailer mounted for towing by truck-tractors or trucks. When connected to an electric power source, the plant shall be capable of self-erection and production. Except for the prime movers and generator sets, the plant

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will be complete and ready for operation as a dry ready-mix plant or as a single and dual mixer wet-mix plant capable of producing pavement and structural quality concrete. When all sections are in operation, the plant will produce 250 cubic yards of concrete per hour. The prime movers and generator sets are not covered by this specification.

3.2 Preproduction model. The supplier shall furnish a plant for examination and test within the time frame specified (see 6.2) to prove prior to starting production, that his production methods and choice of design detail will produce plants that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more plant(s) (see 6.2) for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein and shall be new and unused. Material not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering and production standards of the construction equipment industry. Rubber products furnished shall not exceed two years in age from the date of cure of the rubber to the date of acceptance of the plant by the Government.

3.5 Safety. The safety characteristics of the plant shall meet the following minimum requirements:

(a) All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.
(b) All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel, or which create a fire hazard shall be fully guarded or insulated.
(c) Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment.
(d) Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator.

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(e) All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations.
(f) Live parts of wiring or electrical equipment shall be effectively guarded and grounded to protect all persons or objects from electrical shock hazard.
(g) Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.
(h) All walks and steps shall have nonskid diamond tread perforated surfaces.

3.6 Maintainability. The plant shall operate as specified herein for 500 hours without maintenance other than the manufacturer's recommended normal schedule of adjustments as established by a maintenance schedule prepared and submitted by the supplier prior to test. All assemblies and installed attachments shall be accessible for servicing, repair, and replacement. Sizes and types of connectors, plugs, and valves shall be held to a minimum. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility.

3.7 Design.

3.7.1 Stress limitations. The plant shall be constructed so that the nominal operation and transport static stresses in the load-bearing and load-carrying members shall not exceed 60 percent of the guaranteed yield strength or 30 percent of the guaranteed ultimate strength, whichever is the smaller value. Under dynamic travel conditions with unimproved roads imposing twice the maximum vertical acceleration force, the sum of the static stresses plus the dynamic stresses shall not exceed 100 percent of the yield strength or 50 percent of the ultimate strength, whichever is the smaller value. This analysis shall include calculations and diagrams of the following for each unit furnished:

(a) The load distribution diagram.
(b) The vertical shear diagram.
(c) The bending moment diagram.

3.7.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

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3.7.3 Threaded parts. Threaded parts shall be in accordance with Handbook H28.

3.7.4 Common parts. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, and similar common (standard) parts.

3.7.5 Flexible hose. Except as specified herein, flexible hose, conforming to SAE J40, SAE 40 R2, type F, with matching reusable fittings shall be furnished for airbrake lines, and air pressure control lines.

3.7.6 Breathers. All breathers for lubricant and fluid sumps, brake components, and similar openings on the plant shall prevent the entrance of airborne water and airborne dust.

3.7.7 Liners. All hoppers, bins, chutes, spouts, and other metal surfaces subjected to abrasive wear caused by aggregate movement shall be lined or faced with replaceable high-carbon steel plates with not less than 0.35 percent carbon content. All bolt heads subjected to the flow of material shall not extend beyond the surfaces exposed to the material and shall be of hardened steel.

3.7.8 Drive components.

3.7.8.1 Chain drives. Drive chains, when furnished, shall be in accordance with MIL-C-52058 and shall have provisions for adjustment.

3.7.8.2 V-belt drives. V-belt drives shall be in accordance with USA Standard B55.1 and shall have provisions for adjustment.

3.7.8.3 Gears and gearboxes. All gears shall be enclosed in gearboxes and shall operate in an oil bath. Gears shall conform to USA Standard B6.6, class 2 or 3.

3.7.8.4 Line shafting. All line shafts furnished shall be of carbon steel containing not less than 0.40 percent carbon.

3.7.8.5 Overload safety devices. Replaceable shear collars or the equivalent shall be provided in the drive lines or components throughout the entire mixing plant in all locations where overload may cause damage.

3.7.8.6 Universal joints. All universal joints shall be of the antifriction bearing type.

3.7.8.7 Towed vehicles. Towed vehicles shall conform to the ICC Motor Carrier Safety Regulations, parts 190 and 193, and the the California Vehicle Code, Division 12, with respect to lighting and braking requirements.

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3.8 Capacity. The plant shall produce dry mix concrete of pavement quality, 1-inch slump utilizing 2-1/2- to 3-inch maximum sized aggregate, and structural quality, 3-inch slump utilizing 1-1/2-inch maximum sized aggregate, of the capacity specified in table I.

Table I. Production Capacity

Configuration	Minimum Mixing Time	Minimum Output Volume Per Hour
Dry Batch Ready Mix		100 cu yd
Single Tilting or Horizontal Mixer	1 Min	125 cu yd
Two Tilting or One Tilting and One Horizontal Mixer	1 Min	250 cu yd

3.9 Plant.

3.9.1 Foundation. All foundations shall be incorporated into the structural framework of the major components of the plant as bearing plates or bearing pads. The bearing plates or pads shall be of sufficient square area to permit erection and extended operation with not more than 1-inch settlement on soil having a California Bearing Ratio value of not more than 5 as determined by MIL-STD-621, method 101.

3.9.2 Erection. The plant shall be erected and assembled for complete operation by the use of hydraulic systems constructed as an integral part of each major component, or by means which permit interconnection of components not requiring erection. The use of any external lifting, hoisting, or power equipment not considered as a component of the plant shall not be permitted. Erection and assembly time shall not exceed 8 working hours or 96 manhours.

3.9.3 Compatibility. Without modification, each component of the mixing plant shall assemble and operate in conjunction with other components to form a central plant in all the arrangement described in 3.1. When prepared for travel, the semitrailer-mounted components of the plant shall be compatible for coupling, towing, and maneuvering under all conditions specified herein when coupled to a 5 ton, 6 by 6, truck-tractor conforming to MS500008.

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3.9.4 Weights and dimensions. The gross vehicle weight (GVW) of each major component shall weight of the trailer or semitrailer with all machinery, equipment, accessories, attachments, and lubricants. Component and vehicular ratings shall not be raised to conform to the requirements of this specification. The physical characteristics of each unit prepared for transport shall be as specified in table II, with the unit on level ground and with the inflation pressure of the tires as recommended by the Tire and Rim Association for the imposed GVW loading.

Table II. Weights and Dimensions

Characteristics	Requirements
Overall length including truck-tractor	74 feet, max.
Overall width	132 inches, max.
Overall height	162 inches, max.
Ground clearance, exclusive of suspension	18 inches, min.
Semitrailer:	
GVW	55,000 lb, max.
Fifth-wheel load	15,000 lb, max.
Axle load measured at the ground, each	18,000 lb, max.
Upper fifth-wheel height range	54 to 56 inches
Swing radius clearance	57 inches, max.
Cart-type trailer:	
GVW	14,000 lb, max.
Vertical lunette load, range	10% to 12% of GVW
Axle load, measured at the ground, each	6,000 lb, max.
Tow bar eye height, range	31 to 37 inches, max.

3.9.5 Mixer base section. The mixer base section shall be constructed as a semitrailer. The semitrailer shall be constructed to permit lowering the complete unit to the ground and raising it back to travel position hydraulically. As constructed, the semitrailer shall include provisions for transporting two 100 KW generators during movement of the plant. The base section shall contain the hydraulic systems, connecting arms and tapered pins required to position the horizontal mixer section in its operating position. The bearing plate connecting pad for the tilting mixer section shall be a component part of the mixer base section semi-trailer for operation and transportation.

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3.9.6 Horizontal mixer section. The horizontal mixer section shall be mounted on a semitrailer containing the permanently mounted horizontal mixing drum, electric drive motors, necessary hydraulic system and all required electrical and hydraulic connections and operating controls. The mixing drum shall have replaceable drum liners, mixing blades (or vanes) with replaceable wear plates, wear rings, and drip rings. A charging chute and discharge chute with replaceable wear plates shall also be provided. The drum capacity shall be not less than 4-1/2 cubic yards when rated in accordance with CPMB standards.

3.9.7 Tilting mixer section. The tilting mixer section shall be a semitrailer unit containing the permanently mounted tilting mixer, all necessary linkage, connecting arms, electric drive motor with all necessary cables, connections, a complete hydraulic system, and all operating controls. The mixing drum shall be constructed with replaceable drum liners, replaceable drip rings, and replaceable wear rings as required. The drum capacity shall be not less than 4-1/2 cubic yards when rated in accordance with CPMB standards.

3.9.8 Aggregate section. The aggregate section shall be semitrailer mounted. The section shall have three compartments with a total capacity of not less than 20 cubic yards (30 tons). Clamshell gates shall be provided to discharge material from the compartments to the aggregate batcher unit. The clamshell gates shall be air operated, and solenoid controlled.

3.9.8.1 Aggregate batcher unit. The aggregate batcher unit shall be capable of weighing material for a 4-1/2 cubic yard batch of concrete. The batcher shall be equipped with an aggregate scale and two air-operated solenoid controlled discharge gates.

3.9.8.2 Batching conveyor. The batching conveyor shall have a 36-inch wide endless belt. The conveyor frame shall be of channel-type construction. Troughing rollers, return rollers, head shaft and tail shaft rollers shall be equipped with lubrication fittings. The conveyor shall be electric-motor driven. The conveyor shall be provided with a charging head and appropriate seals to accommodate the horizontal mixer or the tilting mixer. The conveyor shall be covered from the cement entry to the head shaft. A hydraulic system shall be provided to position the conveyor in the operating position. The hydraulic system shall be self-contained within the aggregate section.

3.9.9 Cement section. The cement section shall be semitrailer mounted. The capacity shall be not less than 800 barrels when rated in accordance

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with CPMB standards. Loading of cement shall be accomplished by the transfer of bulk cement from dump trucks to a receiving hopper and then to the cement section by the use of a blower system. A low-pressure aeration system, of sufficient capacity to ensure proper flow of cement, shall be provided in the cement section. Provisions shall be incorporated into the interior construction of the cement section to prevent bridging or consolidation of the cement. Replaceable flow pads shall be provided. The cement section shall deliver loose cement to the discharge assembly either by an auger system or by elevating the cement section hydraulically to an angle not exceeding 30 degrees from the horizontal plane. The auger system or hydraulic system shall be self contained. Discharge of cement from the cement section to the cement batcher unit shall be accomplished by an auger system or a closed conveyor. Power for all operating components shall be by electric motor.

3.9.10 Cement batcher unit. The cement batcher unit shall either be incorporated into the structural frame of the aggregate section or shall be mounted in such a manner as to permit delivery of cement to the charging belt. The unit shall be of appropriate size to batch cement for 4-1/2 cubic yards concrete batches. The unit shall be equipped with a cement scale and an air-operated solenoid controlled discharge gate. Replaceable flow pads and two air vibrators shall be provided.

3.9.11 Cement debagging device. The cement debagging device shall open and unload not less than 1,500 one-cubic-foot paper bags of cement per hour. The cement receiving hopper shall be of sufficient size to receive the cement and shall deliver the bulk cement to the cement section. This device shall also permit the debagging of cement at the plant site.

3.9.12 Air compressor. An electric-driven air compressor of sufficient capacity to operate all components requiring air for operation or activation shall be provided. The air compressor shall be mounted within the structural frame of the aggregate section. The supplier shall provide all gages, lines, fitting, and solenoids necessary to operate the air compressor and air-controlled or actuated units. The air compressor shall be provided with a low-oil-pressure cutoff switch. The cutoff switch shall be wired to a warning light on the control panel.

3.9.13 Water system. The water system shall consist of a 1,000 gallon surge tank constructed as a component of a major component, a water meter capable of metering variable quantities between 35 and 200 gallons, automatic controls, (at the operator's control panel), water pump to supply the mixers, water pump to fill the surge tank, and all necessary lines, piping, fittings, hoses and connections required to fill the surge tank and to supply water through the meter to the mixer. Provisions shall be made for cleaning the water system, and to supply water to truck mixers.

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3.9.14 Air entraining agent system. The air entraining agent system shall consist of a reservoir tank having a capacity of not less than 200 gallons, a transparent graduated tube-type dispenser having a capacity of not less than 430 ounces with a variable metering capability from 128 ounces through 430 ounces, adjustable fill probe and empty signal probe, a manual override control, and all the necessary lines, fittings, piping, valves, vents, and controls necessary to operate the system in conjunction with the water system. The supplier shall provide an electric-driven pump and barrel connections to transfer the liquid add-mixing agent to the reservoir.

3.9.15 Control section. The control section shall be mounted on a cart type trailer. The trailer shall be constructed so as to provide a minimum of 128 square feet of interior floor space. The trailer frame shall be constructed of electrically welded tube and channel. Axles shall be those commerically used for equipment of this nature. Two doors shall be provided, one located on the side and the other at the rear of the trailer. Door opening shall be 32 by 74 inches. The trailer shall be equipped with electric lights wired to the main control panel. A heater and an air conditioner with a cooling capacity of not less than 10,000 Btu/hr shall be provided. Walls and roof shall be insulated with full batt-type fiberglass panels. Exterior shall be prefinished with machine-crimped aluminum having a thickness of not less than 14 gauge. All joints shall be machine locked and caulked. Interior walls and ceiling shall be furnished with plywood panels and painted white. Asphalt tile having a thickness of 1/8-inch shall be provided as flooring. Windows shall provide visibility through a range of 180 degrees. The total window area shall be not less than 60 square feet.

3.9.16 Dump hopper. A dump hopper shall be provided to discharge into dry ready-mix or transit-mix concrete trucks. The hopper shall be of sufficient size to accept a 4-1/2 cubic yard batch of concrete. Discharge shall be controlled by a sliding gate valve or clamshell gate operated from an air solenoid or electric control. The height and width shall permit the free passage of 7 cubic yard truck-mounted mixers. Free fall of mixed concrete shall not exceed 4 feet. A sleeve shall be provided between discharge gate and truck mixer opening.

3.9.17 Scales. Scales shall be of the lever balance or springless dial type and shall conform to CPMB standards. All scales shall include two rows of weight presets and remote scale followers. The aggregate scale shall be the accumulative three material type having a maximum capacity of not less than 18,000 pounds with weighing graduations of not more than 15 pounds, capable of automatically weighing preset amounts of three materials. The cement scale shall be a one material type capable of weighing a preset amount

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of cement. The capacity of the scale shall be not less than 3,000 pounds with graduations of not more than 2-1/2 pounds. The scales shall be equipped with zero interlocks to prevent changing a batch, the sequencency of material of a batch, and discharging a batch prior to reaching the preset weight. Scale accuracy shall comply with CPMB standards.

3.9.18 Electrical system. The power required to operate the complete plant shall not exceed the continuous rated output of four independently connected 100 KW, 240/416 volt, 3 phase, 4-wire generator sets. The generator sets will not be furnished by the supplier.

3.9.18.1 Cable. The plant shall be furnished with four 100-foot cable assemblies conforming to MIL-C-52286, figure 29, four male connector inserts conforming to MIL-C-52286, figure 68, and five pigtails conforming to MIL-C-52286, figure 49, for connecting the generator sets to the main power panel (the extra pigtail is for a spare generator set).

3.9.18.2 Wiring and connectors. Wiring, cable, couplers, plugs, switches, and related components shall be in accordance with NFPA, No. 70, National Electrical Code. The supplier shall provide the plant completely pre-wired to the main power panel. The wiring shall be adequate for the maximum electric power load. Individual loads shall not exceed that specified for any one generator set.

3.9.18.3 Grounding of generator sets and concrete plant. Each separate unit of the concrete plant shall be grounded with two ground rods each. One ground rod shall also be provided for grounding each generator set, including the spare generator set. The ground rods with cable shall conform to MIL-R-11461, type and style optional.

3.9.18.4 Electric motors. All motors required to drive the major components and accessory items of the plant shall be provided. The motors shall be totally enclosed, conform to NEMA MG1, and shall be capable of operating on 416 volt, 60 cycle, 3 phase alternating current (AC). Reduced voltage starters in accordance with NEMA standards shall be supplied.

3.9.18.5 Power panel. The main power panel shall be mounted within the the enclosure on the cart-type trailer. The power panel shall be weather-tight with locking type doors. The power panel shall contain a master switch, fuses, or circuit breakers, and all necessary items for the electrical hookup of the plant. The power panel shall be the central distribution point of electric power for all major components, accessory items, component systems, and operating controls for the plant. All major components, accessory items, and component systems shall be connected directly or through an

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auxiliary power panel to the main control by the use of male-female push pull connectors. Each connector, connector cable, and receptacle shall be coded by matchmarked numbers stamped or permanently affixed to each item.

3.9.18.6 Controls. The control panel shall provide control of all functions of batching, charging, counting, cycle timing, weighing, and slump indications necessary for operation of the plant. The mixer discharge shall be under control of the operator through automatic indication of the mixing time after the slump has been established. All controls shall be interlocked to prevent the mixer charging cycle from actuating unless the mixer and charging belt are running, the air compressor reservoir is at operating pressure, the cement and aggregate batches are complete, the air entraining dispenser is full, and the previously mixed batch is discharged. Interlocks shall also prevent discharge when a batching gate is open or batching when a discharge gate is open. Both mechanical and electric interlocks shall be provided on all air operated gates. Manual override of control systems shall be provided. Indicating lights shall be provided to show that operating systems are functioning properly and the location of any malfunction in regard to the batching and discharge systems.

3.9.19 Hydraulic system. The hydraulic system on the plant shall provide a means of performing all of the various plant functions. The system shall be a 2,000 psi completely closed system and shall include a pump, motor, reservoir, valves, lines, cylinders, fittings, and filter. Oil for the system shall conform to MIL-L-2104, grade 10, or MIL-L-45199, grade 10. All component materials shall be compatible with these oils and shall operate as specified herein throughout an oil operating temperature range of minus 25° to plus 225° F. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100° F. above ambient temperature during all operations specified herein. Unless otherwise specified herein, all system components shall withstand a minimum working pressure of 2,000 psi and a minimum proof pressure of 4,000 psi without permanent deformation or damage, and shall have a minimum burst pressure of 6,000 psi. The system shall perform all functions as specified herein with a maximum pressure equal to relief valve reseat pressure (see 3.9.19.5). All functions shall perform at the rates specified herein when the system full-flow rate is as specified in 3.9.19.2. The oil-pressure drops from pump outlet to pump inlet shall not exceed 200 psi for each of the functions under the following conditions:

(a) Control valves are in center position.
(b) Pump is operating at a speed corresponding to the no-load motor speed.
(c) System oil temperature is 150° plus or minus 5° F.

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Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly of the plant. All pressure, suction, return, pilot, relief, and leakage ports of all hydraulic pumps, valves, actuators, filters, and reservoirs shall conform to SAE J514; O-ring boss for sizes smaller than 1 inch, and SAE J518, 3,000 psi, 4-bolt split-flange for sizes 1 inch and larger. Port sizes shall be as specified in table III and shall limit fluid velocity to 15 feet per second, maximum.

Table III. Hydraulic Pressure Port and Line Sizes

Port Size	Dash No.	Hose ID	Tube O.D.	Tube Wall Thickness
Inch		Inch	Inch	Inch
1/4	-4	1/4	1/4	0.035
	-5	-	5/16	0.049
3/8	-6	3/8	3/8	0.049
1/2	-8	1/2	1/2	0.065
	-10	-	5/8	0.083
3/4	-12	3/4	3/4	0.109
1	-16	1	1	0.120
1-1/4	-20	1-1/4	1-1/4	0.165
1-1/2	-24	1-1/2	1-1/2	0.188
2	-32	2	2	0.250

3.9.19.1 Test points. Steel fittings shall be provided in the hydraulic system for attachment of a hydraulic system tester. One test point fitting shall be in the return line upstream from and adjacent to the filter and the other fitting shall be connected to the outlet of each pump. For line sizes 1 inch and larger, the test point fitting shall have a branch terminating in a 4-bolt, split-flange, port face conforming to SAE J518, 3,000 psi, 1-inch size. For line sizes smaller than 1-inch, the test point fitting shall have

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a branch terminating in an SAE J514, 37-degree flare male, of the same nominal size as the line. The 1-inch, 4-bolt, split-flange fitting shall be sealed with a cap with O-ring. The 37-degree flare fittings shall be sealed with a cap conforming to SAE J514 of applicable size. When a test point sealing cap is removed, the maximum system fluid loss shall be 8 fluid ounces when tested as specified in 4.6.2.1.2. If check valves are used to limit test point fluid loss, the maximum pressure drop across the check valve shall be 5 psi when tested as specified in 4.6.2.1.2. The location of the test point fittings shall be such that:

(a) a tester hose can be attached without the removal of any component other than sealing caps and access panels or plates,
(b) the tester hose can be attached with standard open-end wrenches,
(c) two 10-foot lengths of hose, one attached to the return line test point and the other alternately attached to each pump outlet test point, extend beyond the external surface of the vehicle to a flat horizontal surface on which the connected tester may be placed.

3.9.19.2 Hydraulic pump(s). The hydraulic pump(s) shall be of a constant- or variable-displacement type and shall be gear or shaft driven by the motor. The pump(s) shall have a 2-bolt mounting flange in accordance with SAE J744. At a speed corresponding to the motor operating speed, each pump shall deliver sufficient oil at 2,000 psi and 150° F. plus or minus 5° F. to perform the functions at the rates specified herein. Each pump shall operate continuously without malfunction when tested as specified in 4.6.2.1.3. Each pump shall withstand a minimum proof pressure of 3,000 psi without permanent deformation, damage, or leakage. The overall efficiency of each pump when operating between 750 psi and 2,000 psi, at a speed corresponding to the no-load motor speed and at an oil temperature of 150° F. plus or minus 5° F., shall be a minimum of 75 percent (hydraulic output/mechanical input). There shall be no external leakage such as wetting of the mating portions of the pumps or slinging or dripping of oil from the shafts when tested as specified in 4.6.2.1.3. Each pump shall operate with a minimum inlet port pressure of 7 inches of mercury vacuum corrected to sea level throughout all operating conditions specified herein. The outlet port of the pumps shall be large enough to limit the oil velocity to a maximum of 15 feet per second and shall conform to SAE J518.

3.9.19.3 Hydraulic motor(s). The hydraulic motor(s), if used, shall be of constant or variable displacement type and shall be gear or shaft connected to the driven unit(s). The motor(s) shall have a 2-bolt mounting flange and involute spline in accordance with SAE J744. Each motor, when supplied with

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oil at 2,000 psi and 150° F. plus or minus 5° F., shall perform the functions under all operating conditions specified herein. Each motor shall operate continuously without malfunction when tested as specified in 4.6.2.1.4. Each motor shall withstand a minimum proof pressure of 3,000 psi without permanent deformation, damage, or leakage. The overall efficiency of each motor when operating between 750 psi and 2,000 psi with sufficient oil at a temperature of 150° F. plus or minus 5° F. shall be a minimum of 70 percent (mechanical output/hydraulic input). There shall be no external leakage such as wetting of the mating portions of the motors or slinging or dripping of oil from the shafts when tested as specified in 4.6.2.1.4. Port connections shall be large enough to limit the oil velocity to a maximum of 15 feet per second and shall conform to 3.9.19. Each motor shall operate against a back pressure of 500 psi throughout all operating conditions specified herein.

3.9.19.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall be sealed when the filler cap is in place. No vacuum breaker and pressure relief valves are permitted. The reservoir shall include connectors, baffles, filler opening, screen, cap, and other items required to function as specified. When control valves or the system filter are mounted on or in the reservoir, the mounts and connectors shall meet the requirements specified herein.

3.9.19.4.1 Reservoir mounting. The reservoir shall be mounted to prevent vibrations that may cause leaks, loosening of fasteners, brinelling of material, or cracks. Connections on the reservoir shall be located so that connecting lines can be removed without removing the reservoir or any other major component.

3.9.19.4.2 Reservoir construction. The reservoir shall be located and constructed to operate as specified herein on all slopes and during performance of all functions specified herein. The depth of oil in the reservoir when at the "ADD" mark shall be sufficient to maintain a minimum 4-inch oil depth above the pump suction intake during the performance of all functions and on all slopes specified herein. The pump suction line intake shall be located not less than 3 inches from the bottom of the reservoir. The return lines shall terminate at a minimum of 3 inches below the fluid surface when the fluid level is at the "ADD" mark regardless of the attitude or performance of the plant. The return line termination shall prevent aeration and foaming of the fluid and shall be directed away from the pump suction line. The shortest flow distance between the pump suction intake and return line termination shall be at least 18 inches. The return line shall terminate into a deceleration tube to reduce the velocity of the returning fluid to 4 feet per second maximum. The reservoir shall be constructed of corrosion-

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resistant metal or interior surfaces shall be treated to prevent corrosion. All control linkages and connections into the reservoir shall be sealed to prevent leakage. The reservoir shall be baffled to minimize agitation and aeration of the oil. The hydraulic reservoir filler neck shall have a 3-inch minimum opening and shall contain a removable 100-mesh strainer, retained in place of sufficient size to strain, by gravity, the oil specified herein at 70° F. at a rate of 10 gallons per minute. The filler neck shall consist of a filler tube extending to the "FULL" oil level to prevent overfilling the reservoir. The top of the filler neck shall extend a minimum of 1 inch above the top of the reservoir. The filler neck shall be located to provide for filling from a standard 5-gallon oil container. The filler cap shall be cast metal and shall be permenently marked "Hydraulic System Reservoir" with raised or depressed letters, or by a permanently attached identification plate. Filler neck and cap shall be located to prevent entrance of dust and dirt when removing the cap. The installation and removal of the filler cap shall be by hand without the use of tools. Provision shall be made in the filler cap to insure that the internal air pressure will equalize with atmospheric pressure prior to final disengagement of the filler cap from the filler neck. The filler cap shall be captive chained to the reservoir to prevent loss when it is disengaged from the filler neck. A drain shall be provided to insure complete drainage of the reservoir without disconnecting any line when the plant is on level ground. The drain shall be accessible for removing the drain plug by hand tools and drainage of fluid into a container without spillage on the equipment or the ground. A magnetic device shall be located in the drain plug or a separate connection shall be provided to retain the ferrous metal particles. The device shall be capable of retaining total metal particles equal to 1/200 of the reservoir volumetric oil weight, but not less than 1/2 pound of metal particles. The reservoir shall incorporate a device to indicate the correct fluid level, without opening the reservoir, when the plant is on level ground. The indicator shall display, at a minimum, the "FULL" and "ADD" levels. The sight gage type indicator shall be integrally mounted on a heavy boss for protection against damage and breakage. The indicator shall be so positioned as to be visible for ease of checking and shall be accessible for removal and repair without removal of other parts or components except access cover. A dipstick shall be located in the filler neck, removable without tools, and the "FULL" and "ADD" marks shall be engraved or embossed on one side. The "FULL" and "ADD" marks on the dipstick shall indicate correct oil level when the plant is on level ground, all hydraulic lines are full, and all hydraulic cylinders are retracted. The reservoir shall be provided with an access port(s). The port(s) shall be located and shall be of such size that visual inspection and hand cleaning can be accomplished on all interior surfaces and internally mounted components. Prior to filling the reservoir, all interior surfaces shall be cleaned to remove all traces of contamination. Cleaning

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methods and materials shall be compatible with the hydraulic fluid specified herein. There shall be no visual evidence of contamination when any surface is wiped with a clean, white cloth. After cleaning and prior to installation of attached lines, the reservoir openings shall be sealed to prevent the entrance of any contamination. The reservoir shall withstand a proof pressure of 45 psi without permanent deformation, damage, or leakage when tested as specified in 4.6.2.1.3. The reservoir shall withstand an outside to inside differential pressure of 8 inches of mercury vacuum (4 psi) without evidence of leakage or permanent deformation.

3.9.19.5 Relief valve(s). A hydraulic-system relief valve shall be provided to protect the hydraulic-system from pump pressure exceeding that specified herein. The valve shall operate throughout the temperature ranges as specified herein. The valve shall limit the maximum pump pressure to not exceed 2,000 psi and shall have a reseat pressure of not less than 1,850 psi. The response of the valve shall be such that no pressure peaks shall exceed 3,000 psi. At 1,700 psi pressure and an oil temperature of 150° plus or minus 5° F., the valve when closed shall have a maximum leakage of 25 cubic inches per minute when tested as specified in 4.6.2.1.3. The relief valve shall perform without malfunction when tested as specified in 4.6.2.1.3 and setting shall require no adjustment throughout all tests specified herein.

3.9.19.6 System filter. The system filter shall be the full-flow type and shall have a replaceable element with a 10 micron nominal rating. The filter shall be located so that all return oil shall pass through the filter. The filter shall be located so that the filter element may be removed for examination and replacement without the removal of any components other than the filter cover, and without spilling or draining the oil from the reservoir. Filters located in the reservoir shall be so located as to filter all fluid entering the reservoir through the system return ports, and direction of the filtration shall be from outside to inside the filter element. Filters located outside the reservoir shall be equipped with a drain. Filters shall be furnished with a filter contamination indicator to show when pressure drop across the element exceeds the limits specified herein. At the maximum system oil flow through the filter and an oil temperature of 150° F. plus or minus 5° F., the pressure drop across a new filter shall not exceed 10 psi. An automatic bypass shall be provided in the filter, shall open at not less than 20 psi, and shall permit full flow at a pressure not greater than 30 psi. The filter shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage when tested as specified in 4.6.2.1.3.

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3.9.19.7 Hydraulic control valves. All hydraulic control valves shall be operated by electric solenoids prewired to the control panel located in the control trailer.

3.9.19.8 Cylinders. All hydraulic cylinders shall be double-acting; self alining; and pin, socket or gimbal mounted, and shall be fabricated from seamless steel tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final, machine honing or rolling of the cylinder. Cylinders shall be located so as to provide for ease of maintenance, replacement, and protection from contact with the concrete mix, exposure to the cement dust, sand, rock, and any other material and contact with any vehicle operating around the plant. Any cylinder used for erection shall be protected against dust, rock, water, or any other material entering the cylinder. All erection cylinders shall be used for erection only and shall not be used as a permanent support for the mixing plant during batching operations including raising of the conveyors and any other major component. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have a rod wiper external to the rod seal to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. The rod end cap on all cylinders shall be bolted to the cylinder. All cylinders shall have sufficient strength to perform as specified herein without buckling and bending. Internal flow restriction or bypass shall be provided in the piston cylinder assembly when necessary to prevent damage due to shock loads. The life of the cylinders shall be as follows when tested as specified in 4.6.2.1.5:

(a) Any cylinders continuously used during the batching mixing operation - 750,000 cycles.
(b) Any cylinders used only for erection of the plant - 200 cycles.

3.9.19.8.1 Cylinder tolerances. All hydraulic cylinder bores shall be machine honed or rolled within the limits of plus or minus 0.005 inch of the specified diameter. The cylinder inside diameter (ID) shall be round to within 0.003 inch. The surface roughness of the cylinder ID shall not exceed 32 microinches.

3.9.19.8.2 Cylinder performance. Each cylinder shall have a stroke of sufficient length to position the component as specified herein. Drift of each cylinder piston when supporting a load generating 2,000 psi minimum hydraulic pressure shall not exceed 2 inches per hour at an oil temperature of 150° F. plus or minus 10° F. Rod packing leakage may wet the rod but shall not collect on the rod packing retainer or rod end of the

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cylinder. Rod packing leakage under all conditions specified herein shall not exceed 2.5 grams per 1,000 cycles. The maximum pressure under no-load conditions required to actuate the piston shall be not more than 25 psi on the rod end and not more than 15 psi on the head end. If hydraulic cylinders are used on the tilt mixer they shall incorporate a cushioning device to prevent shocking when the piston is fully extended or retracted.

3.9.19.8.3 Piston and piston rod. Each piston shall have a phenolic (or similar material), bronze, or brass, bearing ring concentric with the piston to within 0.005 inch total indicator reading to carry the piston side loads against the cylinder walls and to prevent contact of the piston outside diameter (OD) and cylinder ID. The surface of the piston rod shall be hardened to 50 to 65 Rockwell C. The rod shall be hard chrome plated. After plating, the maximum rod surface roughness shall not exceed 16 microinches. The plating shall be such that the rod will not corrode when tested for not less than 30 hours as specified in ASTM B117. Any staining which cannot be removed with a oily rag shall be cause for failure.

3.9.19.9 Hydraulic lines. Hydraulic lines shall consist of tubing, hose, and fittings. Hose shall be used where flexing and vibration may contribute to failure. All other lines shall be of tubing as specified herein. Hydraulic lines shall be secured to framing members by cushioned support clamps. All lines shall be routed to provide the shortest line and minimum number of bends consistent with flexibility for fatigue life of the end connection. Line routing shall be simple, provide ease of maintenance and maximum protection. Provisions shall be made to prevent damage of the lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting and shall be supported when necessary to eliminate sagging hose. Pressure hose bend radii shall be not less than that specified in MS39267. All pressure lines shall be located to provide space for installation and maintenance of reusable coupling as specified herein.

3.9.19.9.1 Pressure hose assemblies. The velocity of the oil in all pressure hose shall be not more than 15 feet per second. Hose assemblies in sizes -4 through -12 shall conform to MS39317. Hose assemblies in sizes -16 through -32 shall conform to MS39318. Hose sizes shall be limited to the size specified in table III.

3.9.19.9.2 Suction hose. Suction hose shall be used in the hydraulic suction line between the reservoir and the pump wherever vibration may cause failure of a tube and wherever it may simplify the installation. The suction hose size shall be such that the minimum pump inlet pressure will be not less than 7.0 inches Hg vacuum corrected to sea level. The hose shall

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have a minimum working pressure of 50 psi and a proof pressure of 100 psi. The hose shall withstand an outside pressure of 15 psia with an inside vacuum of 7.0 inches Hg without permanent deformation or damage when tested as specified in 4.6.2.1.6. The change in length of the hose shall not exceed plus 2 percent, minus 4 percent when pressurized from 0.0 psi to 50 psi. The hose shall be installed to eliminate stress concentration and shall be free of kinks.

3.9.19.9.3 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall conform to the requirements specified herein. Tube sizes shall be as specified in table III and the oil velocity in the tubing shall not exceed 36 feet per second. Tubing shall be rigidly clamped to the plant by threaded fasteners spaced at intervals of not more than 24 inches. All bends shall be smooth without flattening, kinking, and wrinkling of the tube. The term "without flattening" is defined as follows: Throughout the bend the maximum diameter minus the minimum diameter shall not exceed 15 percent of the tube nominal diameter. The maximum diameter shall be equal or greater than the nominal diameter. Fittings in sizes -16 through -32 shall be of the 4-bolt, clamp type on the hose end, with an SAE 4-bolt, split flange and stem in accordance with SAE J518 on the opposite end.

3.9.19.9.4 Suction tube and suction tube fittings. Hydraulic suction tube and suction tube fittings shall be used in the hydraulic suction line between the reservoir and the pump where vibration will have no detrimental effect. Tube and tube fittings ID shall be such that the minimum pump inlet pressure will be not less than 7.0 inches Hg vacuum corrected to sea level. The suction tube and suction tube fittings shall have a minimum working pressure of 50 psi and a proof pressure of 100 psi. Tube and fittings shall withstand an outside pressure of 15 psia with an inside vacuum of 7.0 inches Hg without permanent deformation or damage when tested as specified in 4.6.2.1.6.

3.9.19.9.5 Suction hose clamps. Clamps shall be used to secure the suction hose over the suction tube and suction tube fittings. The clamps shall be in accordance with SAE J536, type F, style 1, 2 or 4 and shall conform to the requirements specified herein when tested as specified in 4.6.2.1.6.

3.9.19.9.6 Pressure tube fittings. All pressure tube fittings, when assembled, shall withstand without leakage, damage, or permanent deformation a minimum proof pressure of 4,000 psi and a burst pressure of 6,000 psi. Unless otherwise specified herein, fittings in sizes -4 through -12 shall be in accordance with SAE J514, three piece tube assembly. Tube fittings in sizes -16 through -32 shall contain a female boss into which the tube shall be brazed or welded. The -16 through -32 tube fittings shall terminate either in a flange head which incorporates an SAE J518 4-bolt, split-flange

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mating face, for connection to a hose assembly coupling; or a flange head which integrally incorporates 4-bolt holes and an O-ring groove, for connection to an SAE J518 3,000 psi, 4-bolt split-flange mating port face.

3.9.19.10 Final assembly and cleanliness. All hydraulic components and the hydraulic system shall be kept clean and free from contamination prior to and throughout final assembly on the plant. Prior to acceptance of the plant by the Government, a sample of hydraulic fluid removal from the system reservoir shall have a maximum silting index of 2.0 and shall contain no particles larger than 25 micron. When the plant is tested as specified in 4.6.2.4, there shall be no evidence of any external leakage from the hydraulic system that cannot be corrected.

3.10 Transportability.

3.10.1 Performance characteristics. Each vehicle shall meet the transport performance characteristics specified herein when: (1) prepared for transport; (2) completely serviced and adjusted; and (3) towed by a 5-ton, 6 by 6, military truck-tractor for semitrailers, or by a suitable towing vehicle for cart-type trailers.

3.10.1.1 Highway operation. Each vehicle shall operate over smooth, hard-surfaced primary highways at speeds up to 35 mph.

3.10.1.2 Unimproved road operation. Each vehicle shall operate over unimproved roads at speeds up to 10 mph.

3.10.1.3 Trailing and turning ability. Each vehicle shall trail the towing vehicle without weaving. Each vehicle shall follow the towing vehicle without exceeding 75 percent of the ICC towed-vehicle tolerance. Turns where the angle between the truck-tractor and the semitrailer range up to 90 degrees and between the truck and the cart-trailer and dolly semitrailer combination range up to 60 degrees as measured from straight-ahead to the right and to the left shall not cause damage to or interference between the towed vehicle dolly and the towing vehicle.

3.10.1.4 Braking ability. On each vehicle, the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in accordance with ICC requirements, except that the stopping distance shall be accurately measured from initial movement of the brake pedal to stopping point. When actuated, the emergency breakaway system shall produce smooth emergency brake application. On multiple-axle units, wheel hop during braking shall be prevented.

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3.10.1.4.1 Service brakes. The combined service brakes of the towing and towed vehicles shall control, decelerate, and stop the vehicle combination in accordance with the California Vehicle Code, and ICC regulations (see 3.7.8.7) when loaded within the limits specified in table II. The brake system of the plant component shall be in accordance with TTMA recommended practice RP 3-1-12, except as specified herein. The air leakage within each plant component brake system shall not exceed 0.4 psi per minute with the brakes released, nor exceed 1.67 psi per minute with the brakes applied, each for periods of not less than 10 minutes duration.

3.10.1.4.2 Automatic actuation. The emergency breakaway system shall apply the brakes to stop and hold each plant component, when actuated by disconnect, breakage, or leakage of the emergency line.

3.10.1.4.3 Emergency-parking brakes. The emergency-parking brakes shall function to stop the vehicle combination when the loss of approximately 50 percent of each plant components main airbrake reservoir air pressure has occurred. In addition, the brakes shall provide a positive, continuous braking system, holding the wheels locked for parking of each plant component on grades up to and including 30 percent, despite the depletion of the compressed air supply.

3.10.2 Rail transport. Each vehicle shall withstand the impact force encountered in rail transport without breakage or permanent deformation. The capability to sectionalize each unit shall be provided for rail transportation sectionalization and reassembly of each unit shall be accomplished in not more than 6 hours using only common hand tools and without cutting or welding.

3.11 Vehicular components. The cart-trailer and semitrailer components shall be as specified herein.

3.11.1 Intervehicular connections.

3.11.1.1 Truck-tractor and semitrailer. Each semitrailer shall be furnished with airbrake couplings as specified in 3.11.7 and the electrical intervehicular receptacle as specified in 3.11.9 for connecting the truck-tractor airbrake and electrical systems to the semitrailer systems. The couplings and the receptacle shall be installed on the front of the semitrailer, symmetrical about the centerline in accordance with SAE J702. Clearance about the couplings and the receptacle shall permit connection of the truck-tractor hoses and harness to the semitrailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings

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shall be protected with metal spring dirt covers. The electrical receptacle shall have pigtail leads terminating within a splashproof, semitrailer-type, nosebox. The nosebox shall also house the three circuit breakers as specified in 3.11.9 for the protection of the service-lighting circuits.

3.11.1.2 Truck and cart-trailer. Each trailer, except the control trailer, shall be furnished with an intervehicular electrical harness as specified in 3.11.9, with airbrake lines as specified in 3.11.7 for connecting the trailer to the truck, and with two safety chains. All intervehicular components shall be installed on the trailer so that they shall extend between 36 and 42 inches toward the prime mover from the center of the lunette eye. A plug clip shall be furnished to hold the electrical harness plug and to double the harness back on the trailer tongue. Two dummy couplings shall be furnished and fastened to the tongue for doubling back and holding the lines. The intervehicular line routing, connected and stowed, shall be such that dropping of the tailgate of a military truck will not damage any lines. A semitrailer-type, electrical nosebox shall be furnished.

3.11.2 Frame. The frame shall consist of longitudinal side frame rails reinforced with lateral cross members and gussets to meet the operational conditions specified herein without racking, damage, or permanent deformation. At the supplier's option, each major plant component frame shall be either of bolted frame construction or welded frame construction as specified herein.

3.11.2.1 Bolted frame. The longitudinal frame side rails shall be of heat-treated alloy steel having an ultimate tensile strength of not less than 100,000 psi, an elastic limit of not less than 85,000 psi, and an elongation of not less than 15 percent in 2 inches of length. All junctions and joints of the frame assembly and all component brackets installed on the frame shall be fastened with bolts, washers, and nuts. Welding shall be prohibited. All load carrying bolted connections shall be considered as critical bolted connections (see 3.25.4). Except for composite frames or as recommended by the frame manufacturer, holes shall not be made in the top or bottom flanges of the frame rails. In addition, holes in the frame rail web shall be at least 1-1/2 inches from the top and the bottom flanges.

3.11.2.2 Welded frame. Each major plant component frame shall be alloy steel, channel, tubular, and structural shapes, welded into an integral frame assembly that inherently resists torsional strain and withstands bending, towing, and braking loads. With the exception of removable members and brackets, all junctions and joints of the frame assembly shall be assembled by continuous electric arc welding, using automatic machinery where available.

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3.11.3 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch-diameter lower fifth wheel and shall extend the entire width of the gooseneck. The plate shall be fabricated of high tensile steel with not less than 50,000 psi yield point and shall have a thickness of not less than 0.370 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. An SAE 2-inch-throat-diameter kingpin conforming to MS53036 shall be installed not less than 24 inches or more than 36 inches from the front of the semitrailer. The center of the kingpin shall be within 1/8 inch from the longitudinal centerline of the semitrailer. The swing clearance rearward from the kingpin to any part of the semitrailer below the upper fifth-wheel plate shall be not less than 86 inches.

3.11.4 Cart-trailer towing attachment. Each cart-type trailer shall be equipped with a towing attachment having a ring drawbar coupler conforming to MS51337 installed in accordance with SAE J849. The coupler shall be supported the entire length of the shank in a housing integral with the towing attachment. The clearance between the shank and the housing shall not exceed 0.035 inch. Upper and lower stops shall hold the coupler to prevent rotation. The coupler shall be installed on the towing attachment within 1/8 inch on either side of the centerline of the trailer. The towing attachment may be either fixed or removable type, constructed and installed to provide a structure necessary to resist pulling, bending, and twisting without permanent deformation or damage under all transport conditions specified herein. The towing attachment shall be equipped with tire rub plates on both sides. These plates shall be curved top and bottom to prevent possible interference with the towing vehicle's tire lugs. Two safety chains conforming to ICC regulations shall be furnished. The safety chain hooks shall pass through the shackles of the military truck.

3.11.5 Landing jacks and jacklegs. Landing jacks shall be provided on each semitrailers and jacklegs on cart-type trailers for raising, leveling, and supporting each vehicle. The landing jacks and jacklegs shall be hand-cranked and shall be permenently installed on the vehicle. Operating jacks shall be furnished to raise each vehicle into position for plant operation. The operating jacks shall be permanently attached. All jacks shall be equipped with square, metal, mud pads having a ground bearing area of not less than 144 square inches and shall be interchangeable between all vehicles. Any jack may serve as a combination landing jack or jackleg, and operating jack providing it meets all requirements specified herein. All landing jacks shall withstand the force of a 6-inch drop when the plant is rigged for travel.

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3.11.6 Running gear. The vehicle running gear shall be the sprung type on cart-trailers. The major components shall be of the springless or sprung type on semitrailers. The axle, or the forward axle on tandem or tri-axles, shall be in alinement with the longitudinal centerline of the vehicle so that the two diagonal measurements from the center of the lunette or kingpin to the outer center of each hub do not differ in excess of 1/16 inch. The rear axle of tandem or tri-axle assemblies shall be in alinement with the front tandem or tri-axle so that the parallel measurements between hub centers on each side do not differ in excess of 1/8 inch. Stub axles or through axles may be furnished. The length of axle tubes of through axles may be increased during manufacture providing the mounting pads are increased at least an equal distance to insure that the structural rigidity is equal to that of the axles shown on the standards and drawings. Each cart-type trailer and semitrailer shall be equipped with wheel splash fenders and splash guards which shall restrict water and other road-surface substances thrown from the wheels. These components shall be in accordance with SAE J682, and MS51331.

3.11.6.1 Tandem-axle suspensions. Suspensions shall withstand the imposed loads without overload, permanent deformation, wear, failure, or malfunction. The components and the installation shall be as recommended by the suspension manufacturer for use in severe, abnormal service. The mounting seats shall be installed on the axle in accordance with recommendations of the axle manufacturer. Tandem-axle assemblies shall have divergent angular movement between the front axle and the rear axle. This movement shall permit not less than 8 inches difference in elevation between adjacent wheels on each side as measured from the level vehicular structure. Rubber bumpers shall be provided to cushion the axle at full jounce. For tandems, the suspension configuration shall nullify brake-hop during all degrees of brake application.

3.11.6.2 Tri-axles. Plant components equipped with tri-axles shall use three axles having six single wheels arranged in two lateral rows at the sides with three wheels longitudinally in each row. The axles shall have a rated capacity at least equal to 120 percent of the load imposed on each axle measured at the ground. Wheel bearings shall be of the antifriction, tapered roller-bearing type. Means for fine bearing adjustment shall be provided. Axle spindles shall be hardened and ground.

3.11.6.3 Axle assemblies. The axle assemblies and components of these assemblies are the permissible options to be incorporated in the vehicles of the plant. At the discretion of the supplier, similar standard commercial axles with equal or greater axle manufacturer's load rating are acceptable.

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The selection, quantity of axles, and location shall be at the discretion of the supplier, and the load on any one axle shall not exceed the limit specified in the table II for each type trailer. Cart-type trailer axles shall conform to 3.11.6.3.1. Semitrailer axles shall conform to 3.11.6.3.2 or 3.11.6.3.4.

3.11.6.3.1 Axle assembly - MS35865, type A. The axle shall be interchangeable with respect to bearings and hubs with MS35865, type A. Spindles shall be alloy steel forgings, correctly hardened and ground as recommended by the axle manufacturer. Hub assemblies shall be equipped with MS19081-112 tapered roller bearings, having an inside-to-inside bearing spacing of 4.0 inches, to insure replacement compatibility with hubs of MS35865, type A axles. Bearing cups and cones shall be in matched sets from the same manufacturer. Means shall be provided for fine bearing adjustment. Hubs shall conform to SAE J694, six stud type with 8-3/4-inch diameter bolt circle.

3.11.6.3.2 Axle assembly - MS35865, type C. The axle shall conform to MS35865, type C, dash number optional, except that the straight-air-brake system (see 3.11.7.1) shall be furnished in lieu of the air-over-hydraulic brake system.

3.11.6.3.3 Wheels, rims, and tires. The wheels shall be furnished with wide-base, tubeless tires having traction or rock lug tread design of a size matched to the load and axle in accordance with table II. Tires shall be free of major defects when inspected in accordance with MIL-STD-1224, shall be of not less than 100 level quality, and shall include commercial antiozonant compounds. Tires shall not be loaded in excess of 85 percent of the maximum loads for highway service recommended by TRA. Rims shall be as specified in table IV for the size of tire furnished. Disc wheels shall conform to SAE J876.

Table IV. Tires and Rims

Axle	Tire Size	Ply Rating Minimum	TRA Rim
MS35865, type A	12 x 16.5	10	9.75 x 16.5, 1-1/2 inch negative offset
MS35865, type C	16.5 x 22.5	16	13.00 x 22.5
MS35865, type C	18 x 22.5	16	14.00 x 22.5

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Tires, rims, valves, and caps shall conform to and be installed in accordance with recommendations of TRA. Tire valve caps shall be of metal and equipped with elastomer seals. One assembly of inflated spare tire and wheel, mounted on each semitrailer in a commercial spare tire basket, equipped with locking chain and a padlock, conforming to MS35647, shall be furnished.

3.11.6.3.4 Low-platform semitrailer suspension system. When low-platform semitrailer suspension systems are provided, the axle components shall be interchangeable with the major assemblies shown on MS35866-D1. The maximum axle load shall be limited to 20,000 pounds with dual tires. The brake system shall be of the straight-air type as specified in 3.11.7.1. Commercial wheel assemblies and military tires shall conform to the following:

(a) Wheel assemblies shall be malleable iron or cast steel. Wheels shall be the 15-inch, 6-spoke type with bearing cups pressed in place. Malleable wheels and rim clamps shall be SAE J432, grade 35018. Cast steel wheels shall be SAE J435, grade 0030. Studs and bolts shall be in accordance with SAE J429, grade 5. Brake drums shall be of cast iron SAE J431, grades G3500c or G4000b. The demountable rims shall be 7.5 by 15 size, 5-degree, 3-piece for tube-type tires.
(b) Tires shall be 11.00 by 15, 16-ply rating, conforming to MS35390-47.
(c) Inner tubes shall be 11.00 by 15 conforming to MS35392-26.

3.11.7 Brake system.

3.11.7.1 Airbrake system. Each major plant component (except the control trailer) shall be equipped with a waterproof, internal expanding, straight airbrake system in accordance with ICC Motor Carrier Safety Regulations, section 193.40 through 193.43 and 193.45 through 193.50; and the California Vehicle Code, section 26508, except as otherwise specified herein. The system shall include service and emergency airbrake line couplings, service and emergency line filters, relay emergency valve, air reservoir, fail-safe cylinders or chambers, foundation brake assemblies on each wheel, and all other plumbing and components required for a complete, matched airbrake system. Airbrake line couplings shall conform to MS35746 and to SAE J702, service on the curb side and emergency on the road side, and shall be of the polarized or noninterchangeable types. Each coupling shall be equipped with a spring-loaded dust cover. The couplings shall be color coded blue for service and red for emergency. In addition, line identification tags conforming to MS53007 shall be furnished. The relay emergency valve shall conform to MS53004, except that the rapid bleed-down valve shall be replaced

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with a [illegible] on these straight airbrakes. Conformance of the relay valve to ICC regulations is mandatory. Each air reservoir shall conform to SAE J10. The service brake system shall have a total air storage capacity of not less than 2-times the recommended air volume for truck-trailers in SAE J813. The reservoir shall be equipped with a remote control or a 90-degree twist drain valve mounted on the low end of the reservoir and positioned on the curb side of each plant component, accessible to personnel from the standing or kneeling position. The size of the foundation brakes on each axle shall be the same and the swept drum total braking surface of each axle (2 spindles) shall be not less than 330 square inches. The foundation brake components shall not be overstressed or exceed the brake component manufacturer's ratings when a full service brake application is performed. All components of the brake system shall be located, installed, and protected to prevent damage or malfunction from water, ground obstructions, and debris, and to obtain maximum efficiency and ease of maintenance. No part of the brake system shall extend below the bottom of the rim of the wheels. The brake lines shall be routed within the protection of the vehicle framing and shall be secured by cushioned support clamps. Securing shall be required at intervals not to exceed 3 feet. Grommets or through-frame connectors shall be installed where lines pass through sheet metal and framing members. The ends of these lines shall be identified by markers conforming to MS39020 installed near the fittings. The brake lines components shall conform to the following:

(a) Flexible air line - MIL-H-3992 type I.
(b) Rigid air line - SAE J844 copper tubing.
(c) Fittings - SAE J512, heavy-duty, airbrake compression type, reusable.

At the suppliers option, SAE J40, SAE 40R2, type F, air hose may be furnished in lieu of (a) and (b) above.

3.11.7.2 Inertia hydraulic brake system. The control trailer shall be equipped with an inertia brake system. The inertia brake system shall consist of a main housing, master cylinder, tow-bar assembly, all lines, fittings, wheel cylinders, brake shoes, backing plates, anchors, springs and all other required items to complete the brake system. The actuator shall be mounted on the tow bar. The housing shall be internally threaded at both ends. The master cylinder shall contain all the elements of a conventional automotive master cylinder, plus a separate accumulator chamber. The master cylinder shall be threaded into the rear end of the housing. The master cylinder bore and piston shall be assembled on a common centerline

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with the drawbar plunger. The compression stroke of the tow bar to the master cylinder shall be transmitted by a direct acting push rod. A pre-loaded coil spring shall be mounted inside the actuator housing and shall function as a hold-off to prevent premature, intermittent and undesirable braking snubs. The brake system shall provided for emergency actuation when accidental uncoupling from the towing vehicle occurs during travel.

3.11.7.3 Brakes. The brake lining shall conform to HH-L-361. All metal surfaces enclosed by the brake drum, including the interior surface of the drum, metal surfaces of the shoes, surfaces of backing plates, and all attached parts, shall be coated with primer conforming to MIL-P-46093 or TT-P-664. Ramps, wedges, pins, eccentrics, cam levers, anchors, and retracting springs shall be coated with preservative oil conforming to MIL-L-21260, grade 2. Items or surfaces which are wholly prime coated, painted in accordance with Government specifications, or constructed of materials providing corrosion protection equivalent to that provided by MIL-C-16173, grade 1, do not require additional protection.

3.11.8 Chock blocks and cradles. Two wood chock blocks and two wire cradles shall be furnished for each axle assembly. The cradles shall be located near the wheels, on the front of the fenders, when practicable. The chock blocks shall be captive chained to the cradles on the outer side with chains of sufficient length to permit placement of the chock blocks either in front of or behind the vehicle's wheels. The chock blocks shall be as shown in figure 1 and the cradles shall be as shown in figure 2.

3.11.9 Vehicular electrical system. Each major plant component and cart-type trailer shall be equipped with a military, waterproof, 24 volt vehicular electrical system for service and blackout night travel. Except as specified herein, the electrical system shall be in accordance with the following requirements:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b) ICC Motor Carrier Safety Regulations.
(c) TIMA Recommended Practice RP 6-2-9.
(d) ATA Specification E-3.
(e) Figure 3.

All components and wiring shall be so located and guarded as to provide protection from vibration, breakage, and damage from falling and flying aggregates. The electrical components shall be in accordance with table V. Item quantities that are not specified shall be furnished as required (AR) for each major plant component and cart-type trailer.

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Table V. Vehicular Electrical System

Description	Requirement	Quantity Per End Item
Service lights: clearance, sidemarker, and identification.	MS35423	14 - under 30 ft. 16 - 30 ft & over
Blackout lights; clearance, sidemarker.	MS35424	8
Service combination lights: Stop, turn signal, and tail.	4-in. dia waterproof, double contact	2
Blackout combination lights.	MS51330	2
Reflectors.	MS35387	8
Intervehicular harness (for cart trailers only).	MIL-C-13486/12, length as specified with plugs conforming to MS75020 on each end, mating towing vehicle to unit	1
Front trailer receptacle.	MS75021-1 with matching cover	1
Low-tension cable.	MIL-C-13486, 14 AWG minimum	AR
Harness insulation.	MIL-I-15126 & MIL-I-631	AR
Loom.	SAE J562 oil resistant	AR
Cushioned support clamps.	MS21322, MS21333 & MS21334	AR
Circuit number bands.	MS39020	AR
Friction connectors, waterproof type.	MS27142 thru MS27148 and MS33800	AR
Connector lubricant.	MIL-S-8660	AR
Connector holder clip.	Plastic or metal, spring clip type	1 each per connector
Threaded plugs and receptacle.	Military waterproof type	AR
Terminal connectors.	Spade type with insulator sleeve (SAE J561) with heat shrinkable tubing	AR
Circuit breakers.	MS39062	3

3.11.9.1 Lighting. All lights shall be installed on the semitrailers and trailers and shall operate as specified herein on 24 volts. Shock and vibration protection shall be provided for all lights to prolong filament

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life. Both clearance lights and sidemarker lights shall be provided on all four corners. Identification, sidemarker, and clearance lights shall be clustered and mounted on plates inserted into lamp bracket sockets conforming to SAE J725. The two service combination lights shall be sealed-beam units mounted in elastomer rings, or bulb, socket, and lens sealed in elastomer housings and shall be the flush- or recess-mounted type. The two blackout combination lights shall be installed inboard of the service combination lights. These two blackout lights shall be installed flush within 6-inch-minimum-diameter holes in the rear apron or crossmember or shall protrude not more than 4 inches rearward; and shall be protected with guards, equal to 6-inch-diameter, schedule 60, steel pipe. The rear of the four combination lights shall be boxed and shielded to prevent damage in any environment specified herein. Blackout clearance and sidemarker lights shall be positioned below the corresponding service clearance and sidemarker lights.

3.11.9.2 Wiring. The wiring for both service and blackout lights shall be as shown on figure 3. Grommets shall be used to prevent chafing where wire and harness pass through any structural member. Loom shall be furnished to protect wires and harness from road splash. Cushioned support straps and loom shall prevent whipping, chafing, or other mechanical damage to the wires and harness. Splices shall be insulated and waterproofed. Two or more wires following the same routing shall be formed into harnesses with either one-half overlapping turns of 0.010-inch-thick, 1-inch wide tape or with insulation. All wires shall be number coded near each connection. Friction connectors shall be lubricated with connector lubricant prior to assembly and each assembly shall be secured to the semitrailer or cart-type trailer with a connector holder clip. Waterproof electrical, semitrailer-type nosebox shall be furnished. The nosebox shall contain terminal connectors for all of the live and the two ground circuits and shall house the three circuit breakers. The nosebox shall be mounted on the front of the trailer. From the nosebox, one main harness shall be routed rearward along the frame sidemember to the rear lights and incorporate branch harnesses routed to the side lights.

3.11.9.3 Reflectors. Reflectors shall be mounted in accordance with ICC requirements. In addition to ICC requirements, two amber reflectors shall be mounted on the front of the vehicle at approximately the same height as those mounted on the sides.

3.11.10 Bearings. Unless otherwise specified herein, all bearings shall be greaseable antifriction type conforming to FF-B-171, FF-B-185, or FF-B-187, as applicable. Mountings of bearings shall permit split tapered sleeve and locking nut or inner race secured to the shaft by a minimum of two set screws.

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3.11.11 Servicing and adjusting. Prior to delivery of the vehicles, the supplier shall adjust and service each vehicle for travel use including at least the following: Adjust brake system; check electrical system; inflate all tires; and completely lubricate the chassis and components as specified in 3.13.

3.12 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the plant shall conform to MIL-STD-461, class IIB.

3.13 Lubrication. All surfaces requiring lubrication shall be provided with suitable means for lubricating.

3.13.1 Lubricants. The plant shall operate as specified herein when lubricated with military lubricants (see 6.6).

3.13.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, and III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.13.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.13.4 Filling, draining, and checking devices. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a drain plug conforming to MS35844 or MS49006 and with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the plant is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts except that panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed.

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3.13.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827 as applicable. The mixing plant shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases may not be compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.13.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F. to 32° F. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 except that 1 percent copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish may be used in lieu of salicylanilide, the specified fungitoxicant:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces which have been previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.15 Lifting attachments. Each separate unit and each sectionalized component shall be provided with lifting attachments. The attachments shall enable the unit or component to be lifted in its normal traveling or operating position. The attachments shall be located so that not less than 1-inch clearance is maintained between the slings and all exterior parts of the item being lifted. Spreader bars are permitted. Each attachment shall withstand not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load when such limitation is possible. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the

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attachment. The inside diameter of the eye shall be not less than 3 inches. Members to which attachments are fastened and the fasteners shall withstand all stresses imposed by the attachment. Where required, structural members shall be reinforced to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.16.

3.16 Tiedown attachments. Each separate unit and each sectionalized component shall be provided with integral attachments to permit tiedown to the floor or deck of the transportation medium. Each attachment shall withstand, without damage to the plant, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals cargo weight of the unit or component, as applicable. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.15.

3.17 Marking.

3.17.1 Identification. Each separate unit of the plant shall be identified in accordance with MIL-STD-130. The marking shall be applied to each separate unit in plates conforming to MIL-P-514, type I, composition C of type I, grade A, class 1 material. The plate shall include the nomenclature of the complete plant as well as of the unit to which affixed.

3.17.2 Shipping data. Shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and, in addition, shall show the silhouette of the applicable unit in transport position indicating the center of gravity, and the location and capacity of the lifting and tiedown attachments.

3.17.3 Instruction plates. Each separate unit of the plant shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the plant. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material. The words "Warning", "Caution", and "High Voltage" shall be in red matching FED. STD. No. 595, color No. 31136.

3.18 Treatment and painting. The plant components shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, semi-gloss, color No. X24087, in accordance with FED. STD. No. 595.

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3.19 Vehicle marking. The registration numbers and other marking of the separate units shall conform to MIL-STD-642. The GVW shall be marked on each forward side of the unit. The prescribed tire pressures shall be marked on the unit near the tires. All marking, except for tire pressure, shall be white reflectorized material conforming to L-S-300, type and class optional.

3.20 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

3.21 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.22 Toolbox(s). Each major plant component shall be provided with a weatherproof toolbox made of steel not less than 0.071 inch thick. The toolbox shall be provided with a lid having a continuous hinge along one edge. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two hook-type tension fasteners shall be provided to keep the lid closed. Not less than a 3/4-inch clearance shall be provided between the hook lever and any other part of the toolbox. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock conforming to MS35647 shall be provided for locking the toolbox. The toolbox shall be fastened to each major component in a protected position. A drain hole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the tools furnished.

3.23 Items to be furnished with the plant. No shipment of plants shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the plant without such items has been received from the contracting officer.

3.24 Government loaned property. The following property in the quantities indicated will be loaned by the Government for testing as specified in 4.6.2:

Item	Description	Identification	Quantity
1	Generator Set, Skid-Mtd 100 KW, 3 Phase, 60 Cycles	FSN 6115-752-1588 MIL-STD-633	4
2	Truck, Tractor, 5 ton, 6 x 6, M52	MS500,008	2

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3.25 Workmanship. All workmanship on each plant shall be in accordance with engineering, manufacturing, and production and safety standards for the manufacture of equipment of this nature. Workmanship deficiencies, if any, shall be discovered and corrected. All parts, components, and assemblies of the plant, including bearings, seals, machinery, stampings, welded parts, castings, forgings, and machine work shall be clean and free from sand, dirt, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs and sharp edges and corners, except where sharp edges are functional.

3.25.1 Aluminum and steel fabrication. Aluminum and steel used in the fabrication of the plant shall provide original quality surface finish and shall be free from kinks and sharp bends. Aluminum and steel having eroded surfaces is not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be free of burrs, slag, and sharp edges. Precautions shall be taken to avoid overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends of a major character shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

3.25.2 Welders and welding.

3.25.2.1 Welders qualification. Welder qualification shall be in accordance with MIL-W-52574.

3.25.2.2 Welding. Welding of steel shall be in accordance with MIL-W-52574, welding process optional, except that joint welding procedures are not required for type 1 steels.

3.25.2.3 Aluminum welding. Aluminum welding shall be accomplished by the metal-inert-gas method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; however, hammered welds shall not be acceptable. Preheating for welding is permissible, providing the temperature does not exceed 400° F. for a total time of 30 minutes. Aluminum weld processess requiring the use of welding flux shall not be used.

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3.25.3 Brazing. Brazing of steels, nickel alloys and copper alloys shall conform to MIL-B-7883. Brazing of aluminum and aluminum alloys shall not be permitted.

3.25.4 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Washers or lockwashers, shall be provided. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All capscrews, bolts, and nuts shall be tight and correctly torqued. Critical load carrying bolted connections (see 3.11.2.1) shall, additionally, be as follows: Bolt holes shall be reamed as required. All junctures and joints shall be fastened with high strength bolts with washers and nuts, hex or 12-point type. Bolts shall be not less than SAE J429, grade 5 and may be of higher grade as necessary. Washers shall be of the flat type, sized to spread the clamping loads such that brinnelling, or embedding in the component members shall be prevented. Mating nuts shall be of the self-locking type having elastic collars, including nuts, selected from MIL-N-25027. Fasteners shall be lubricated prior to wrenching and installation tools shall be proofed at set intervals to insure constant and correct torque performance.

3.25.5 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.25.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.25.7 Forgings. Forged components shall be in accordance with the Forging Industry Handbook.

3.25.8 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The supplier is responsible for insuring that components and materials used are manufactured examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. Prior to testing, the plant shall be examined for the defects marked "X" in column 1 of table VI. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Hydraulic components specified in 3.9.19 shall be subjected to the tests marked "X" in column 1 of table VII. Upon successful completion of the examination specified in 4.3.1, the plant shall be subjected to the tests marked "X" in column 2 of table VII. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. An initial production plant(s) will be selected at random by the Government from the first 2 plants being produced from production tooling and will be examined for the defects marked "X" in column 2 of table VI and subjected to the tests marked "X" in column 3 of table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production plant shall not exclude the remaining plants from the quality conformance inspection and acceptance provisions specified in section 4. Failure of an initial production plant to meet any requirement specified herein during and as a result of the examination and tests specified shall be cause for rejection of the initial production plant(s) and shall be cause for refusal by the Government to continue acceptance of production plants until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no

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cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all plants accepted prior to completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of all tests specified in 4.5.2, each plant shall be examined for the defects marked "X" in column 3 of table VI. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each plant shall be subjected to the tests marked "X" in column 4 of table VII in the following order:

(a) Cleanliness (see 4.6.2.1.8).
(b) Compatibility (see 4.6.2.4).

Failure of either test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table VI.

Table VI. Examination Schedule

Preproduction	Initial Production	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4	5
X	X	-	101. Material.	3.4
X	X	-	102. Safety characteristics.	3.5
X	X	-	103. Maintainability.	3.6
X	X	-	104. Stress limitations.	3.7.1
X	X	-	105. Dissimilar metals.	3.7.2
X	X	-	106. Threaded parts.	3.7.3
X	X	-	107. Liners.	3.7.7
X	X	-	108. Drive components.	3.7.8
X	X	-	109. Foundation and erection.	3.9.1, 3.9.2

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Table VI. Examination Schedule (Cont'd)

Preproduction	Initial Production	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4	5
X	X	-	110. Compatibility.	3.9.3
X	X	X	111. Weights and dimensions.	3.9.4
X	X	-	112. Mixer base section.	3.9.5
X	X	-	113. Horizontal mixer section.	3.9.6
X	X	-	114. Tilting mixer section.	3.9.7
X	X	-	115. Aggregate section.	3.9.8
X	X	-	116. Cement section.	3.9.9
X	X	-	117. Cement batcher.	3.9.10
X	X	-	118. Cement debagging device.	3.9.11
X	X	-	119. Air compressor.	3.9.12
X	X	X	120. Water system.	3.9.13
X	X	X	121. Air entraining agent system.	3.9.14
X	X	-	122. Control section.	3.9.15
X	X	-	123. Dump hopper.	3.9.16
X	X	X	124. Scales.	3.9.17
X	X	-	125. Hydraulic test points.	3.9.19.1
X	X	-	126. Hydraulic pump.	3.9.19.2
X	X	-	127. Hydraulic motor.	3.9.19.3
X	X	-	128. Reservoir.	3.9.19.4
X	X	-	129. Relief valves.	3.9.19.5
X	X	-	130. Filter system.	3.9.19.6
X	X	-	131. Hydraulic control valves.	3.9.19.7
X	X	-	132. Cylinders.	3.9.19.8
X	X	-	133. Hydraulic lines.	3.9.19.9
X	X	-	134. Transportability.	3.10
X	X	-	135. Intervehicular connections.	3.11.1
X	X	-	136. Chassis frames.	3.11.2
X	X	-	137. Semitrailer upper fifth-wheel and kingpin	3.11.3

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Table VI. Examination Schedule (Cont'd)

Preproduction 1	Initial Production 2	Quality Conformance 3	Defect 4	Requirement Paragraph 5
X	X	X	138. Cart-trailer towing attachment.	3.11.4
X	X	-	139. Landing jacks and jacklegs.	3.11.5
X	X	-	140. Running gear.	3.11.6
X	X	-	141. Brake systems.	3.11.7
X	X	X	142. Chock block and cradle.	3.11.8
X	X	X	143. Vehicular electrical system.	3.11.9
X	X	-	144. Bearings.	3.11.[illegible]
X	X	X	145. Servicing and adjustment.	3.1[illegible]
X	X	X	146. Electromagnetic interference not identical to preproduction model.	3.[illegible]
X	X	X	147. Lubrication, lubricants and components.	3.13 thru 3.13.6
X	X	X	148. Fungus and moisture resistance.	3.14
X	X	-	149. Lifting and tiedown attachments.	3.15, 3.16
X	X	X	150. Marking.	3.17
X	X	X	151. Treatment and painting.	3.18
X	X	X	152. Vehicle marking.	3.19
X	X	X	153. Technical publications.	3.20
X	X	X	154. Repair parts and tools.	3.21
X	X	-	155. Toolbox.	3.22
X	X	X	156. Items to be furnished.	3.23
X	X	X	157. Workmanship.	3.25

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Table VII. Test Schedule

Preproduction	Production					
Hydraulic Components	Plant	Initial Production	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	-	-	-	Hydraulic system.	4.6.2.1.1	3.9.19, 3.9.19.5, and 3.9.19.8.2
X	-	-	-	Test point connections.	4.6.2.1.2	3.9.19.1
X	-	-	-	Hydraulic pump, reservoir, relief valve and filter.	4.6.2.1.3	3.9.19.2, 3.9.19.4.2, 3.9.19.6 and 3.9.19.15
X	-	-	-	Hydraulic motor.	4.6.2.1.4	3.9.19.3
X	-	-	-	Hydraulic cylinder components.	4.6.2.1.5	3.9.19.8 and 3.9.19.8.2
X	-	-	-	Hydraulic suction hose.	4.6.2.1.6	3.9.19.9.2
-	X	X	X	Cleanliness.	4.6.2.1.7	3.9.19.10
-	X	X	-	Road.	4.6.2.2.1	3.10
-	X	X	-	Brake.	4.6.2.2.2	3.10
-	X	X	-	Landing gear.	4.6.2.2.3	3.11.5
-	X	X	-	Tandem axle articulation.	4.6.2.2.4	3.11.6.1
-	X	X	-	Scales.	4.6.2.3.1	3.9.17
-	X	X	-	Aggregate section capacity.	4.6.2.3.2	3.9.8
-	X	X	-	Aggregate batcher unit.	4.6.2.3.3	3.9.8.1
-	X	X	-	Cement section.	4.6.2.3.4	3.9.9
-	X	X	-	Cement batcher section.	4.6.2.3.5	3.9.10
-	X	X	-	Water system.	4.6.2.3.6	3.9.13
-	X	X	-	Water metering system.	4.6.2.3.7	3.9.13
-	X	X	-	Ad-mix tank.	4.6.2.3.8	3.9.14
-	X	X	-	Ad-mix dispenser.	4.6.2.3.9	3.9.14
-	X	-	-	Productive capacity.	4.6.2.3.10	3.8
-	-	X	X	Compatibility test.	4.6.2.4	3.9.3
-	X	X	-	Lifting attachments.	4.6.2.5	3.15
-	X	X	-	Tiedown attachments.	4.6.2.6	3.16

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4.6.2 Test and test conditions. Prior to being tested, the plant shall be lubricated with oils and greases as specified in 3.13. Oils shall be those designed for use in the ambient temperature at which the tests are being performed. The water tank shall be filled to the capacity specified in 3.9.13, except for the test specified in 4.6.2.2. The air entraining agent system reservoir tanks are to be filled to the capacity specified in 3.9.14, except for the test specified in 4.6.2.2. The air compressor shall be operated during operating phase of test and the pressure of the reservoir tank maintained at the manufacturer's specified pressure. All hydraulic reservoir tanks shall be filled to the specified level. Generator sets shall operate at the prescribed voltage and cycles. During operating phases of the plant tests the electric motors shall be operated at their normal operating speed. Wood blocking as required shall be provided by the supplier. Soil analysis shall indicate the CBR Value of the soil on which the plant is erected (see 3.9.1). As service and adjustment are performed the mixing plant shall be inspected for ease of maintenance. Nonconformance to 3.6 shall constitute failure of this test. Test as indicated in 4.6.2.2 shall be conducted first. Test conditions for 4.6.2.2 shall consist of servicing the plant for travel. Water and air entraining agent tanks shall be empty during tests specified in 4.6.2.2.

4.6.2.1 Hydraulic systems and components.

4.6.2.1.1 Hydraulic system. The hydraulic system shall be tested in accordance with test method No. 1. Evidence of contamination, a pressure drop in either direction exceeding that specified in 3.9.19, a relief valve pressure setting not as specified in 3.9.19.5, a drift rate exceeding the cumulative rate permitted by the drift rate specified in 3.9.19.8.2, malfunction or binding of any cylinder, excessive silting index or particle size as specified in 3.9.19.10, or evidence of leakage that cannot be stopped as specified in 2(d) of the test method shall constitute failure of this test.

4.6.2.1.2 Test point connections. The hydraulic test point connections shall be tested in accordance with test method No. 2. Excessive system fluid loss from any one test point, inability to attach tester hose, or excessive pressure drop across test point check valve when compared to 3.9.19.1 shall constitute failure of this test.

4.6.2.1.3 Hydraulic pump, reservoir, relief valve and filter. The hydraulic pumps, reservoirs, relief valves, and filters shall be tested in accordance with test method No. 3. Except as specified herein, the above

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components shall be endurance tested concurrently for a total of 1,000 hours. Tests under each set of conditions specified in the test method shall be continuous and shall be conducted in the following sequence:

(1) Performance and proof pressure test (concurrent testing not required).
(2) Endurance tests.
(3) Performance tests (repeat of (1)).
(4) Examination.

Evidence of any of the following shall constitute failure of this test:

(a) Reservoir - Any external leakage, permanent deformation or damage, capacity less than that specified in 3.9.19.4.2, air entering the pump, peeling and chipping of internal surfaces, or contamination.
(b) Pump - Any external leakage, galling, scoring, efficiency less than specified, inability to meet proof pressure, inability to complete the test, or contamination (see 3.9.19.2).
(c) Relief valve - Pressure limits not as specified, pressure peak in excess of that specified, any leakage, scuffing, galling, or contamination (see 3.9.19.5).
(d) Filter - Pressure drop in excess of that specified, deterioration, bypass opening pressure in excess of that specified, inability to meet proof pressure specified or contamination (see 3.9.19.6).

4.6.2.1.4 Hydraulic motor. The hydraulic motor(s) shall be tested in accordance with test method No. 4. Failure to operate continuously, inability to meet proof pressure, efficiency less than specified, external leakage of oil, galling, scoring, or inability to complete the test shall constitute failure of this test (see 3.9.19.3).

4.6.2.1.5 Hydraulic cylinder component. One cylinder from each different hydraulic operation shall be tested in accordance with test method No. 5, test procedure 2, a, b, c, e and f. For tests on the tilter and transfer gate cylinder, if used, the test shall be in accordance with test method No. 5, 2, d in addition to 2, a, b, c, e and f. The flow rate shall be equal to that produced at no-load motor speed at the above pressure. Evidence of external or internal oil leakage in excess of that specified in 3.9.19.8.2, evidence of any breakage or permanent deformation, excessive pressure required to move the rod end of the piston and the head of the piston, inability of the cylinders to operate as specified in 3.9.19.8, excessive wear of any component, or evidence of contamination, galling, scuffing, or piston-to-cylinder contact as specified in test method No. 5 shall constitute failure of this test.

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4.6.2.1.6 Hydraulic suction line. The reservoir-to-pump suction line assembly shall be tested in accordance with test method No. 6. A change in length of the hose in excess of that specified in 3.9.19.9.2, evidence of defects or deterioration of the hose, evidence of leakage, or evidence of slippage of the hose on the tube shall constitute failure of this test.

4.6.2.1.7 Cleanliness. Upon completion of plant tests as specified in 4.6.2.4, remove a sample of hydraulic fluid from each hydraulic reservoir, perform the test specified in test method No. 1 and 2. Evidence of contamination which cannot be corrected as specified in test method, or leakage that cannot be corrected by adjustment, shall constitute failure of this test.

4.6.2.2 Vehicle performance test.

4.6.2.2.1 Road test. The separate units shall be towed with a truck tractor as specified in 3.10.1 for the distances as specified in table VIII.

Table VIII. Road Test

Terrain	Speed	Distance
Highway operation.	35	75 miles
Unimproved gravel roads.	10	50 miles

(a) Measure and record the deviation or off tracking of semitrailer with respect to the prime mover, when the semitrailer is towed straight ahead on a paved surface.
(b) Angle the vehicle combination as specified in 3.10.1.3.
(c) Turn the prime mover, towing the semitrailer or trailer, in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction.

Failure to complete the test, inability to operate and function as specified herein, or nonconformance to 3.10 shall constitute failure of this test.

4.6.2.2.2 Brake test. Test the towed vehicle for conformance to ICC Motor Carrier Regulations, the California Vehicle Code and 3.10. Tests shall be conducted with rated payload, unless different loading is specified.

(a) Tow the unit over the test course and, when it reaches a speed of not less than 20 mph, signal the driver to make a panic brake stop using both prime mover and towed vehicle brakes. Perform the test four times, two times in each direction, allowing not more than 2 minutes between tests.

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(b) Tow the unit at not less than 35 mph and apply the service brakes to produce a deceleration rate of 10 feet per second per second, until the unit has slowed to 10 mph. Then immediately accelerate at wide-open-throttle again up to not less than 35 mph. Repeat the snubbing cycle for a total of 5 cycles and make a panic brake stop on the 6th cycle.

(c) Tow the unit over the test course and, when it reaches 20 mph, cause the towed unit to go into "emergency", simulating a trailer breakaway from the prime mover. Restore the towed unit brake system to normal. Disconnect the emergency line hose and observe if there is an air leak at the airbrake line coupling.

(d) With the unit stopped on a paved surface, uncouple the prime mover brake lines from the towed unit and bleed down all towed vehicle air reservoirs. Determine by the drawbar pull method or on actual grades the maximum holding ability of the towed unit brakes. Release the emergency parking brakes using only tools supplied with each vehicle and without resupply of compressed air.

(e) Test the vehicle airbrake system as specified in TIMA recommended practice RP 3-1-12 and in 3.8.10.4.

(f) Test air leakage as specified in 3.10.1.4.1.

Failure to complete the test, inability to function and operate as specified or nonconformance to 3.8.1.4 through 3.8.1.4.3 shall constitute failure of this test.

4.6.2.2.3 Landing gear test. Couple the semitrailer to the truck-tractor. Adjust the landing gear legs to provide not less than 6-inches clearance height above a concrete surface. Set the semitrailer brakes into emergency, and rapidly drive the truck-tractor out from under the semitrailer causing the semitrailer to fall 6-inches onto its landing gear prop legs. Operate the landing gear between limits of travel and inspect for damage. Evidence of malfunction, binding, or permanent deformation or nonconformance to 3.11.5 shall constitute failure of this test.

4.6.2.2.4 Tandem-axle-assembly articulation. When tandem axles are used, arrange the vehicle in a level position on two blocks, one per side, 8 inches high. Evidence of binding, frame interference, or loss of tire-ground contact shall constitute failure of this test (see 3.11.6.1).

4.6.2.3 Plant component and capacity test.

4.6.2.3.1 Scales. Calibrate the aggregate scales using test weights. Weigh 5,000, ~~10,000~~ 15,000, and 18,000 pounds of material. Reweigh the material on a certified scale. Weigh three materials accumulatively to a total of 15,000 pounds. Reweigh the material on a certified scale. Inability of the scale to function properly, variation of accuracy in excess plus or minus 2 percent, or nonconformance to 3.9.17 shall constitute

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failure of this test. Calibrate the cement scale using test weights. Weigh 1,000 and 2,500 pounds of material. Reweigh material on a certified scale. Inability of the scale to function properly, accuracy variation in excess of plus or minus 1 percent, or nonconformance to 3.9.17 shall constitute failure of this test.

4.6.2.3.2 Aggregate section capacity. Determine the capacity of the aggregate bins by weight or volumetric method. Failure of the aggregate section to meet the requirement as specified in 3.9.8 shall constitute failure of this test.

4.6.2.3.3 Aggregate batcher unit. Determine the capacity of the aggregate batcher unit using the weight method. Discharge the batcher using the clam-shell doors. Nonconformance to 3.9.8.1 shall constitute failure of this test.

4.6.2.3.4 Cement section. Determine the capacity of the cement section by filling the trailer as specified in 3.9.9. Failure of the cement section to accept the quantity of cement as specified in 3.9.9 shall constitute failure of this test.

4.6.2.3.5 Cement batcher section. Fill the cement batcher as specified in 3.9.10 observing the operation of the scales. Discharge load using discharge doors. Operate air vibrators to expedite the discharge. Failure of the batcher to empty completely shall constitute failure of this test.

4.6.2.3.6 Water system. Determine the capacity of the water tank by metering water into the tank as specified in 3.9.13. Failure of the tank to meet the capacity specified in 3.9.13 shall constitute failure of this test.

4.6.2.3.7 Water metering system. Test the metering capability of the water metering system by metering 50, 100, 150, and 200 gallons of water into a container and weigh each sample. A tolerance of one percent plus or minus shall be acceptable. Failure of the system to meter the water accurately within the specified tolerance or nonconformance to 3.9.13 shall constitute failure of this test.

4.6.2.3.8 Air entraining agent system tank. Determine the capacity of the air entraining agent system tank by metering. Failure of the tank to meet the capacity specified in 3.9.14 shall constitute failure of this test.

4.6.2.3.9 Mixture dispenser. Meter 128 ounces and 430 ounces of mixture through the system into a separate container of known volume or weight. Compute the quantity of material discharged from the system. Measured volume not within plus or minus 3 percent of indicated volume shall constitute failure of this test (see 3.9.14).

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4.6.2.3.10 Production capacity test. Erect the plant within the time frame specified in 3.9.2. Operate the plant in its various configuration. Inability of the plant to produce concrete within the volume ranges as specified in table I shall constitute failure of this test.

4.6.2.4 Compatibility test. Erect each plant (except preproduction model) using each configuration, within the time frame as specified in 3.9.2. Interconnect each component and physically operate each component in conjunction and in sequence with the other components of the plant (see 3.9.3). Calibrate each scale and operate each scale using test weights. Use electric power provided by Government-furnished 100 KW generator sets. Excessive settlement of the plant into the soil, inability to erect and interconnect components in all configurations within the time frame as specified in 3.9.2, or malfunction of the scale, water system, air entraining system, hydraulic systems, air system, or any accessory component shall constitute failure of this test.

4.6.2.5 Lifting attachment. Each separate unit shall be suspended by the lifting attachments with multiple slings. The load on each lifting attachment shall be determined with each unit in suspension. Each unit shall be lowered and each lifting attachment shall be subjected to a pull equal to or greater than 2-1/2 times the load reading previously obtained on the same lifting attachment while each unit was suspended. Nonconformance to 3.15 shall constitute failure of this test.

4.6.2.6 Tiedown device. The empty weight of each separate unit shall be determined. Each tiedown device shall be subjected to the proportionate share of the forward, rearward, upward, and sideward thrusts specified in 3.14. Thrust measurements shall be made with a dynamometer or other accurate measuring device. Nonconformance to 3.16 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select plants or components at any time during the contract production period and subject these items to the examination specified in 4.6.1 and to the tests specified in 4.6.2 through 4.6.2.6 to determine that the quality of the selected plants is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Plants or components will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production plants. Failure of any plant may be considered cause for refusal by the Government to continue acceptance of plants until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

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4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.1. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of crated components for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Examination. The preservation, packaging, packing, and marking shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

158. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
159. Sectionalization and disassembly not as specified for level A or B.
160. Parts not matchmarked to facilitate reassembly for level A or B.
161. Cloth tags for matchmarking and lubrication information not waterproofed for level A.
162. Matchmarking and lubrication information tags not attached for level A or B.
163. Unpainted exterior metal surfaces not coated with preservative for level A or B.
164. Exposed exterior machined surfaces not coated with preservative, wrapped, or covered with barrier material, and barrier material not secured as specified for level A or B.
165. Trailer and prime mover components not preserved in accordance with the referenced document for level A or B.
166. Electric motors not preserved in accordance with the referenced document for level A or B.
167. Gears not preserved as specified for level A or B.
168. Drive chains not preserved as specified for level A or B.

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169. Shafts and universal joints not coated with preservative as specified for level A or B.
170. Drive belts not removed or released from tension for level A or B.
171. Unpainted pulley grooves not coated with primer as specified for level A.
172. Belt fasteners not coated with varnish as specified for level A.
173. Conveyor belt not released from tension for level A or B.
174. Conveyor belt, when removed, not compactly rolled and secured as specified for level A or B.
175. Air compressor and accessories not preserved and packaged in accordance with the referenced document for level A or B.
176. Water system not completely drained.
177. Water pumps not preserved as specified for level A or B.
178. Hydraulic system not preserved as specified for level A or B.
179. Scales not locked or blocked to prevent movement.
180. Components of the scales requiring a contact preservative not coated and covered or wrapped as specified for level A or B.
181. Instruments not protected as specified for level A.
182. Electrical component openings not sealed as specified for level A.
183. Repair parts not preserved in accordance with the referenced document.
184. Maintenance tools not preserved and packaged in accordance with the referenced document for level A or B.
185. Intermediate packaging not as specified for level A or B.
186. Toolbox lids not secured to prevent pilferage for level A or C.
187. Disassembled components or intermediate containers placed on the units in a manner that would increase cubage, or interfere with lifting or mobility for level A.
188. Blocking, bracing, and anchoring of crated components not in accordance with the referenced document for level A.
189. Depreservation guides not prepared, placed in waterproof envelopes, and secured to the basic unit for which intended.
190. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packaging, packing, and marking complies with the preparation for delivery requirements of this specification (see 6.4). Examination shall be as specified herein.

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5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. The plant shall be sectionalized to meet the requirements of 3.10.2. In addition, components of units and sections vulnerable to damage, pilferage, and loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools, shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Removed keys shall be secured in the keyway of the pulley or flywheel.

5.2.1.2 Matchmarking. To facilitate reassembly, removed components and sections shall be matchmarked. Large sections and components shall be matchmarked by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121 or MIL-V-1174; or matchmarked with paint conforming to MIL-P-13983. Small removed components shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. The cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

5.2.1.4.1 Unfinished surfaces. Components of the plant such as tie rods, adjusting rods, clevises, coil springs, shafts, couplings, liners, blower system, auger system, brackets, framing, leaf springs, linkages, ratchets, pawls, grease fittings, washers, pins, bolts, nuts, or any other similar unpainted exterior metal surface not definitely specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Unpainted metal identification and instruction plates shall be coated with varnish conforming to MIL-V-13811.

5.2.1.4.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to PPP-T-60, type I.

5.2.1.5 Trailer and prime mover components. The components of each trailer, and semitrailer, shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281.

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5.2.1.6 Electric motors. Each generator shall be preserved in accordance with the level A requirements of MIL-E-16298.

5.2.1.7 Gears.

5.2.1.7.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.7.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10, grade 1, 2, or 3 preservative as applicable and tagged to indicate: "The preservative is good for operation. Do not drain until the first required lubrication change". Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged to indicate: "The lubricant is good for operation". Gears not operating on lubricating oil (SAE 10, 30 or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable-type lubricant". Tags shall conform to UU-T-81, type A. Tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.8 Drive chains.

5.2.1.8.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained, and the entire chain shall be coated with type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with soft, annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, type I. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.8.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.7.2 for enclosed gears and gear housings.

5.2.1.9 Shafts and universal joints. Exposed surfaces of shafts and universal joints shall be coated with type P-1 preservative.

5.2.1.10 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

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5.2.1.11 Conveyor belts and fasteners. Metal belt fasteners shall be coated with varnish conforming to MIL-V-1174 or TT-V-121. The belt shall be removed or released from tension. Removed belts shall be compactly rolled, and the roll shall be secured with strapping conforming to QQ-S-781, type I, class B, size as applicable. The straps shall be placed circumferentially on the roll and tensioned so as not to damage the belt. Unpainted surfaces of the pulleys and rollers shall be coated with primer conforming to TT-P-664.

5.2.1.12 Air compressor. The air compressor and accessories shall be preserved and packaged in accordance with the level A requirements of MIL-C-3600.

5.2.1.13 Water system. The complete water system shall be completely drained and all openings sealed with caps, plugs, or tape conforming to PPP-T-60, type I. In addition, the pumps shall be preserved by fogging type P-10, grade 2 preservative through the pump intake in sufficient quantity to assure coating all interior surfaces and parts, while the pump is being slowly actuated.

5.2.1.14 Hydraulic systems (except hydraulic brake or steering systems). The hydraulic fluid supply tanks shall be filled with the hydraulic fluid recommended for operation. The pistons shall be retracted as far as practicable into the cylinder and secured. When the pistons cannot be fully retracted, exposed portions of the piston rods (ramshafts) shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2; extending the wrap approximately 2 inches on the ram cylinder, if possible. The wrap shall be secured in place with tape conforming to PPP-T-60, type I. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position. Hose shall not be disconnected. Any unpainted exterior metal surfaces of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. A tag conforming to UU-T-81, type A shall be attached to the control lever indicating: "The hydraulic supply tank is filled with the fluid required for operation". The tag and printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.15 Scales. The scales shall be locked or blocked to prevent movement. Surfaces of components requiring a contact preservative shall be coated with type P-7 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and the wrap secured with tape conforming to PPP-T-60, type I.

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5.2.1.16 Instruments. Instruments not protected with a cover or housing shall have the dial glasses of instruments or the complete panel covered with a fitted piece of 1/4-inch thick plywood or with hardboard and secured in place with tape conforming to PPP-T-60, type I.

5.2.1.17 Electrical component openings. Openings into the starter switches, controls, junction boxes, and power receptacles shall be sealed with tape conforming to PPP-T-60, type I.

5.2.1.18 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.19 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of MIL-H-15424 and placed in the toolbox.

5.2.1.20 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1, and placed in the toolbox.

5.2.1.21 Other components. Other components not specifically mentioned herein shall be preserved as specified for similar components specified herein.

5.2.1.22 Intermediate packaging. Packaged and unpackaged components, accessories, and repair parts that cannot be secured to the basic units shall be intermediate packaged in the toolboxes as space permits or in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional.

5.2.2 Level B. The complete plant shall be preserved and packaged as specified for level A, with the following exceptions.

5.2.2.1 Tags. Cloth tags specified in 5.2.1.2, 5.2.1.7.2, and 5.2.1.14 shall not require waterproofing.

5.2.2.2 Trailer and prime mover. The components of each trailer and each prime mover shall be preserved and packaged in accordance with the level B (mobile) requirements of MIL-STD-281.

5.2.2.3 Exposed chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 or P-9 preservative shall be applied.

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5.2.2.4 Drive belts. Drive belts shall be released from tension.

5.2.2.5 Conveyor belt fasteners. Belt fasteners shall not be required to be coated with varnish.

5.2.2.6 Electrical component openings. Openings shall not be required to be sealed with tape.

5.2.2.7 Intermediate packaging. Intermediate packaging shall be as specified for level A, except boxes shall be domestic type or class as applicable.

5.2.3 Level C.

5.2.3.1 Repair parts, maintenance tools, and technical publications. Repair parts and technical publications shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation but shall be placed in the toolbox as specified for level A.

5.2.3.2 Water system. The water system shall be drained and openings sealed as specified in 5.2.1.13.

5.2.3.3 Scales. The scales shall be locked and blocked to prevent movement.

5.2.3.4 Other components. Preservation and packaging of all other components of the mixing plant shall afford protection against deterioration and damage from the supplier to the initial destination.

5.3 Packing. Packing shall be level A or C, as specified (see 6.2).

5.3.1 Level A. The mixing plant shall be prepared for mobile shipment. The toolbox lids shall be closed and secured to prevent pilferage. When sufficient space is available, packaged and unpackaged components required for each unit or section shall be located on the unit or section for which the components are intended. The components shall be placed and secured in any available space that will not increase cubage or interfere with lifting or mobility of the units. The components shall be secured to the units with zinc-coated strapping conforming to QQ-S-781 or QQ-S-790. Packaged and unpackaged components that cannot be secured to the units or to the sections shall be packed in crates conforming to PPP-C-650, style A, type optional; MIL-C-3774, nailed assembly, skid-type base; or MIL-C-104, type I, class 1 or

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2, style A or B, as applicable. Blocking, bracing, and anchoring shall be in accordance with the applicable crate specification and with MIL-STD-1186. Crated components shall be shipped at the same time as the unboxed (mobile) units.

5.3.2 Level C. The complete mixing plant shall be prepared for unboxed (mobile) shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rating. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When sufficient space is available, packaged and unpackaged components required for each unit or section shall be placed and secured on the unit or section for which intended in any available space that will not interfere with lifting or mobility of the units. Packaged and unpackaged components that cannot be secured to the units or sections shall be packed in crates complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules. Crated components shall be shipped at the same time as the unboxed (mobile) units.

5.4 Depreservation guide. Depreservation guides shall be prepared, placed in waterproof envelopes marked "Depreservation Guide", and secured in a conspicuous location on each basic unit. Unless otherwise specified (see 6.2) and 6.8), DA Form 2258 - Depreservation Guide for Vehicles and Equipment shall be used.

5.5 Marking. Packages, shipping containers, and unboxed (mobile) units shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The mixing plant is intended for use as a complete central mixing concrete plant intended to produce quality structural and pavement concrete. The plant is designed and intended to be used as a dry batch ready mix plant, a wet batch transit mix plant, or a single or dual drum central mixing plant. The plant will require supplemental equipment for operation. This supplemental equipment, to be provided by the Government, includes the following:

(a) Generator sets, 100 KW, 120/208-240/416 volts, 60 cycle. Four required for operation plus one standby set.
(b) Truck, tractor, 5 ton, 6 by 6, M52 MS500008-1 for each semitrailer.
(c) Trucks or truck-tractors suitable for towing trailers.

When the plant is destined for immediate use, the contracting officer should arrange for obtaining the supplemental equipment.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When an initial production mixing plant is required (see 3.3).
(d) Number of mixing plants to be furnished for initial production inspection when applicable (see 3.3).
(e) Technical publications required (see 3.20).
(f) Applicable list of repair parts and maintenance tools (see 3.21).
(g) Time frame required for submission of preproduction pack (see 5.1).
(h) Level of preservation and for packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the mixing plants will be held in covered storage.
(i) When other than DA Form 2258 is to be used (see 5.4).

6.3 Preproduction model. Any changes or deviations of production plants from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier from his obligations to furnish plants conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of the production packs from the approved preproduction packs will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligations to preserve, package, pack, and mark the plants in accordance with this specification.

6.5 Government-loan property. The contracting officer should arrange to loan the property specified in 3.24.

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the mixing plant covered by this specification as contained in Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.7 DA Form 2258. The contracting officer shall arrange to furnish DA Form 2258 when requested by the supplier (see 5.4).

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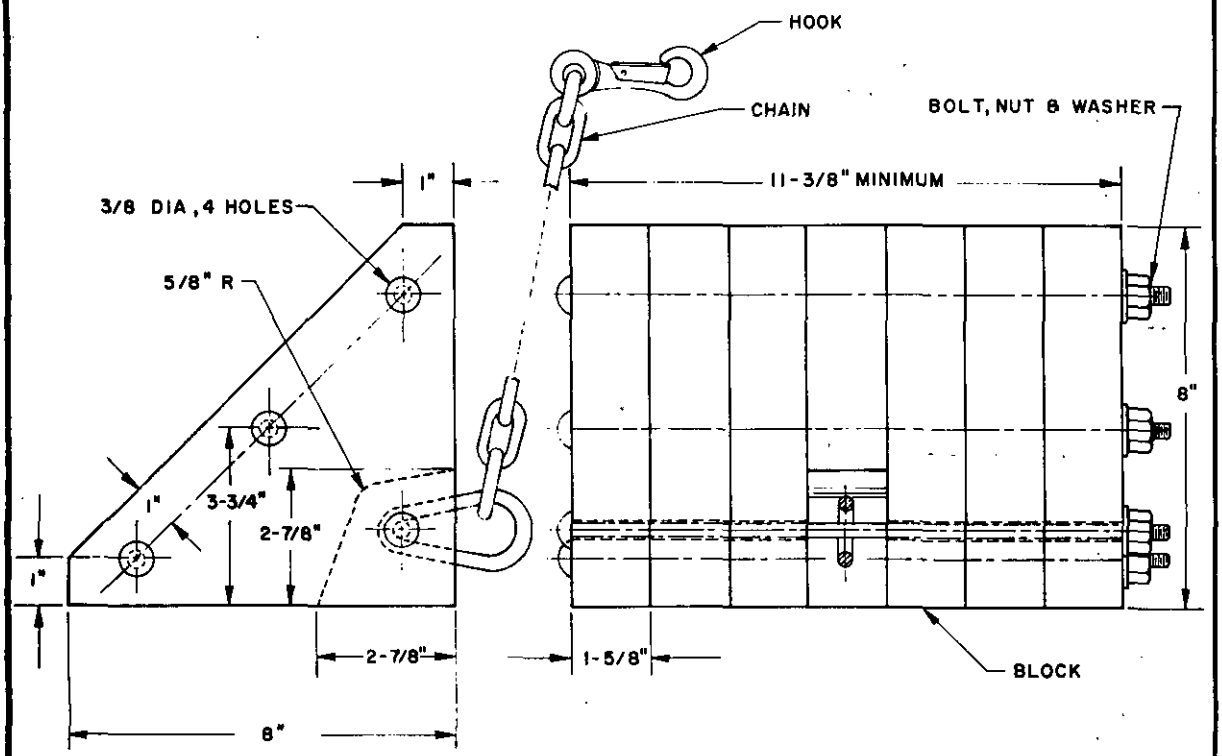
Army - ME

Preparing activity:

Army - ME

Project No. 3895-A066

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT/100 FT. = 70 LBS.

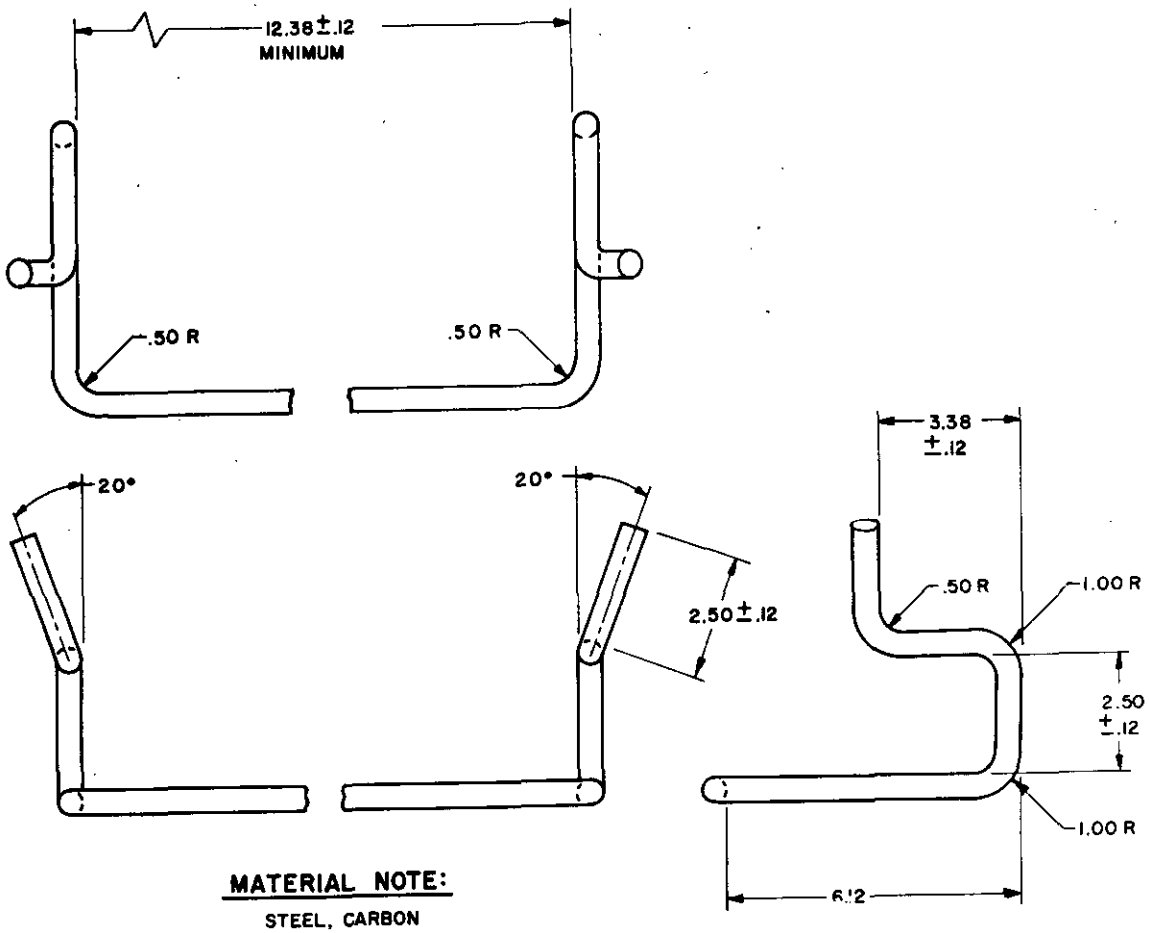
MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 1. CHOCK BLOCK ASSEMBLY

DX-603

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MATERIAL NOTE:

STEEL, CARBON
SAE 1015 TO SAE 1025
1/2 DIA

FIGURE 2. CRADLE

DX-604

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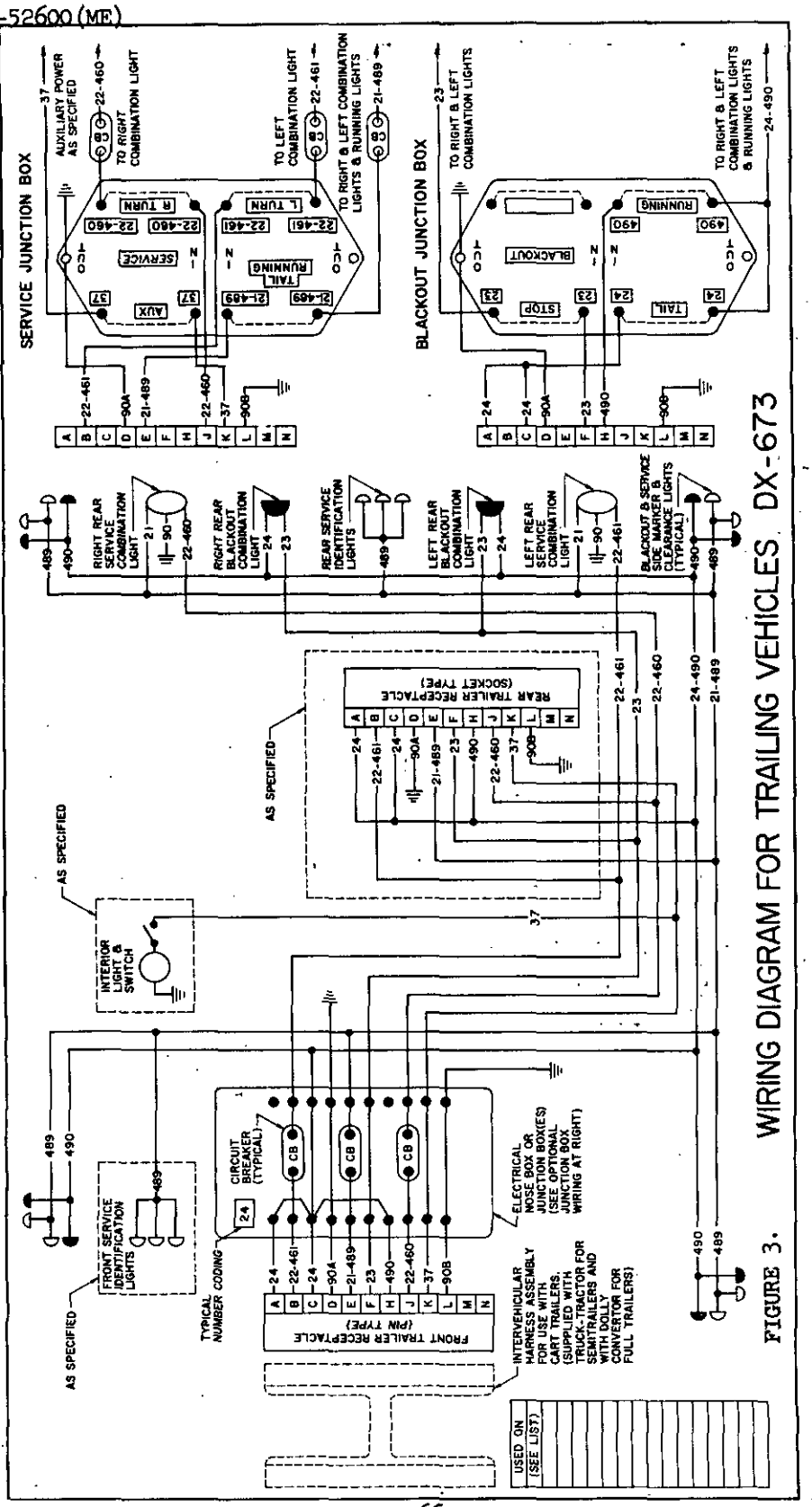


FIGURE 3. WIRING DIAGRAM FOR TRAILING VEHICLES DX-673

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TEST METHOD NO. 1

HYDRAULIC SYSTEM TEST

1. TEST APPARATUS:

(a) Pressure gages.
(b) Temperature gages.
(c) Timing device accurate to 0.10 second.

2. TEST PROCEDURE:

(a) Connect the hydraulic lines at the hydraulically driven actuators to bypass related actuators. Install a new filter element in the plant system.
(b) Run the motor at no-load speed pumping the oil at 150° plus or minus 5° F. through each circuit for a minimum of 5 minutes in one direction for each function. Record the pressure drop. Repeat the above, reversing the flow of each function. Repeat with the controls in the center position.
(c) Operate each function of the system for 15 minutes continuously through full stroke and at maximum rates specified. During operation of attachments, record the rate of travel.
(d) With the system oil temperature at 150° plus or minus 5° F., position each control lever and pressurize each pressure line of the system to relief pressure and hold for a minimum of 15 seconds. Examine for external leakage. Leakage due to defective seals and loose bolts may be corrected by replacement of the seals and tightening of the bolts to the torque recommended by the supplier providing no machining or welding is required. After repair of the leaks has been made, repeat the leakage test.
(e) Repeat (d) above at no-load governed engine speed and record the maximum stabilized pressure of each relief valve.
(f) Remove hydraulic fluid sample from the reservoir and determine the silting index in accordance with procedure specified in ASTM F52. Examine filter removed from silting index equipment for evidence of particles exceeding specified size.

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TEST METHOD NO. 2

TEST POINT CONNECTING TEST

1. TEST APPARATUS:

   (a) Pressure gages.
   (b) 10-foot hoses as required for test points.
   (c) Weighing scales.
   (d) Additional items to complete the testing.

2. TEST PROCEDURE:

   (a) Remove the test point sealing caps and measure and record the fluid loss from each check point.
   (b) Attach a 10-foot length of hose between the return line test point and a system tester and a second 10-foot length of hose between each pump outlet test point and tester.
   (c) Remove test point check valve, if used, from the test point. Apply oil pressure to the valve at a maximum rate of 0.5 psi per second and determine and record the valve "cracking" pressure.
   (d) Pass oil through the valve at full system volume and determine and record the pressure drop across each valve.

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TEST METHOD NO. 3

PUMP, RESERVOIR, RELIEF VALVE, AND FILTER TESTS

1. TEST APPARATUS:*

(a) Tilting test bench with the following:

(1) Relief valve - 3,000 psi pressure minimum.
(2) Control valve.
(3) Temperature gages and recorders.
(4) Pressure gages and recorders.
(5) Torquemeter.
(6) Rpm meter.
(7) Flowmeter.
(8) Timer.
(9) Cycle counter.
(10) Hourmeter.
(11) Oil cooler.
(12) Transparent suction tube.
(13) Additional items to complete the system assembly.

(b) Oil conforming to MIL-L-2104, grade 10.

2. TEST PROCEDURE:

(a) Pump.

(1) Install the pump in the test system as shown in figure A-1 or on a dynamometer test bench. Measure and record the overall efficiency (hydraulic output/mechanical input) at the speed corresponding to motor no-load speed at an oil temperature of 150° plus or minus 5° F. and at a minimum of 6 pressure points at equal increments starting with 750 and continuing through 2,000 psi.
(2) Increase the pump pressure to 3,000 psi maintaining an oil temperature of 150° F. plus or minus 5° F. Hold for 30 seconds and check for external leaks at the assembled surfaces and the shaft seal.
(3) Install the system for running the endurance test and cycle as follows: 5 seconds at minimum pressure and 5 seconds with the pressure limited to not less than 2,000 psi for a total of 1,000 hours. Examine for aeration in the transparent pump suction tube at least once every 24 hours of operation. For the first 50 hours, the oil temperature shall be at 215° plus or minus 5° F. The oil temperature at the pump inlet shall be 150° plus or minus 5° F. for the remaining 950 hours of the test.

*All instruments shall have an accuracy of plus or minus 2 percent.

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(4) Repeat 2(a) (1).
(5) After completing the foregoing tests, disassemble and examine the pump.

(b) Reservoir.

(1) Close all outlets of the reservoir, fill to "FULL" mark on the dipstick and record the capacity.
(2) Completely fill the reservoir with oil and pressurize to the working pressure defined in 3.9.19.4.2 and hold for 30 seconds. Examine for leaks, permanent deformation, or damage.
(3) Install in the reservoir a transparent window 8 inches wide from below the oil suction opening to 2 inches above the maximum oil level so that when lighted, the suction opening will be visible. Place the reservoir in the circuit as shown in figure A-1 and fill the reservoir to the "ADD" mark on the dipstick. The reservoir shall be oriented to the identical location in relation to the pump as it is on the plant. The reservoir shall be tilted to an angle which will be identical to that when the plant is on a 15 percent slope in the direction which is most critical for pump suction. Increase the system oil temperature to 215° plus or minus 5° F. with the fill cap removed. Seal the reservoir by replacing the cap. With the pump operating under pressure, drain a volumn of oil equivalent to the rod displacement of all cylinders on the plant from the pressure line and reseal the line. Measure and record the reservoir and the pump inlet pressures. Maintain these conditions throughout the first 50-hour portion of the test in 2(a) (3). Upon completion of the first 50 hours, remove the fill cap, replace the equivalent amount of drained oil, and reseal the reservoir by replacing the fill cap.
(4) Run the remaining 950 hours of the endurance test with the reservoir-pump installed in its position when the plant is on level ground.
(5) Disassemble the reservoir and examine for contamination and condition of internal parts.

(c) Relief valve.

(1) Apply oil pressure to the relief valve in the relief range at a maximum rate of 1 psi per second. Determine and record the pressure at which the valve "cracks" open and the pressure at full flow. Reduce oil pressure through the relief valve at a maximum rate of 1 psi per second, and determine and record

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when the valve reseats. Measure and record the leakage with the oil at 150° plus or minus 5° F., and the pressure at 1,600 psi plus or minus 10 psi.

(2) The system relief valve shall be the load device for the pump test (see 2(a) (3)) for a minimum of 150 hours at 150° plus or minus 5° F., and for 50 hours at 215° plus or minus 5° F. For the remainder of the pump test, an alternate load device may be used.

(3) Repeat 2(c) (1) upon completion of 2(c) (2).

(4) Disassemble the valve and examine for signs of scuffing, galling, and contamination.

(d) Filter.

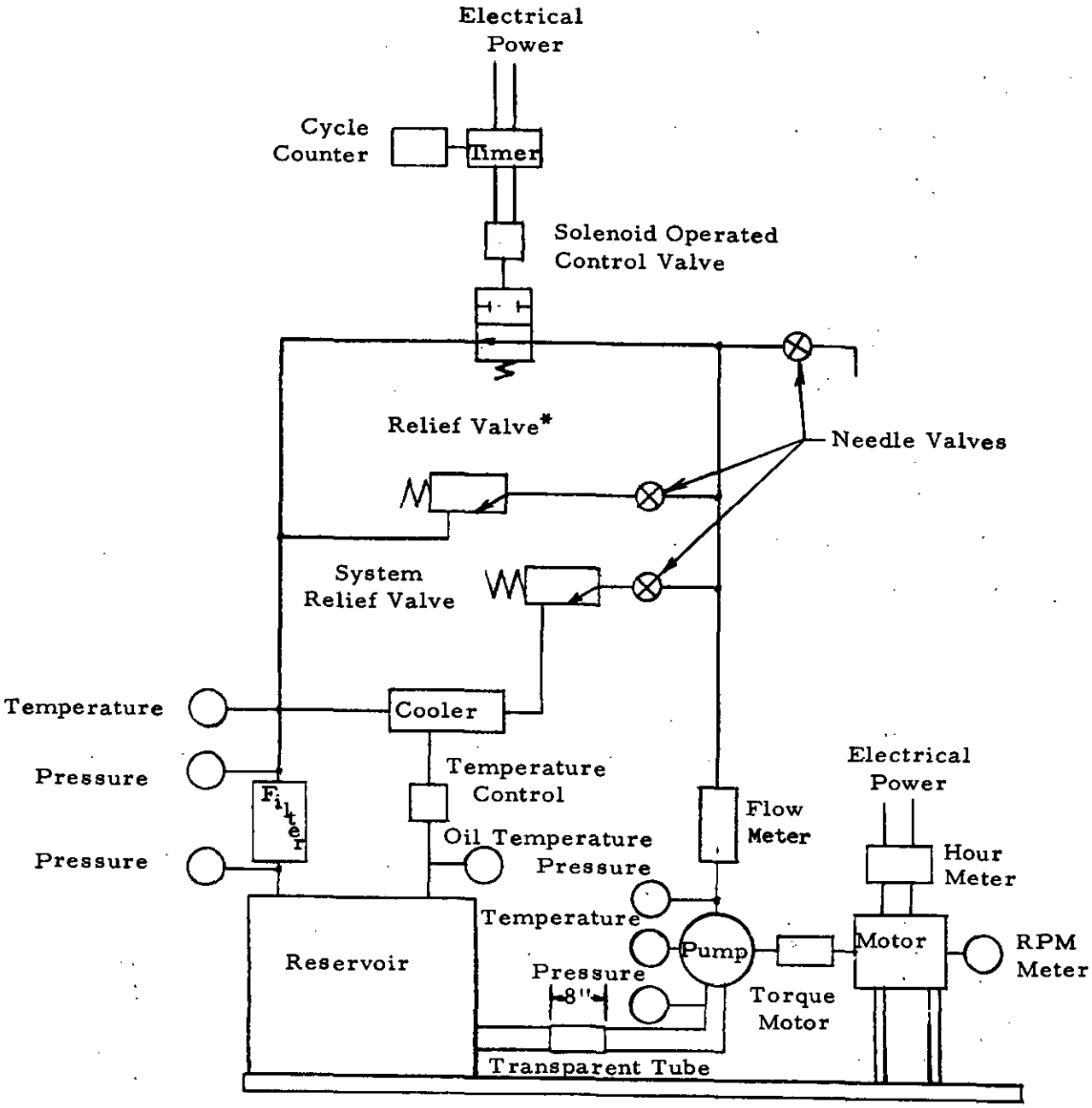
(1) Pressurize the filter to 100 psi and hold for 30 seconds. Examine for leaks.

(2) Install the filter with a new element in the system as shown in figure A-1. Measure and record the oil pressure drop across the filter at maximum system flow and at an oil temperature of 150° plus or minus 5° F.

(3) Plug the filter element chamber oil exit, measure and record pressure drops across the bypass valve at minimum and full flow with the oil temperature at 150° plus or minus 5° F.

(4) Measure and record the pressure drop across the filter upon completion of each 500 hours of endurance testing. Remove and spread out the filter element and examine for the degree of contamination and breakdown. Compare with an unused element.

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Note: Relief Valve(*) setting at 150% of max. system working pressure.
All instruments shall have an accuracy of ±2 percent.

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FIGURE A-1 Test Circuit Diagram for Pump, Reservoir, Relief Valve and Filter Tests

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TEST METHOD NO. 4

HYDRAULIC MOTOR TEST

1. TEST APPARATUS

(a) Test bench with the following:

(1) Relief Valve.
(2) Control valve - same as used on crane.
(3) Temperature gages and recorders.
(4) Torquemeter.
(5) Rpm meter.
(6) Flowmeter.
(7) Timer.
(8) Cycle counter.
(9) Hourmeter.
(10) Oil cooler.
(11) Dynamometer.
(12) Additional items to complete the system assembly.

(b) Oil conforming to MIL-L-2104, grade 10.

2. TEST PROCEDURE

(a) Install the motor in the test system.
(b) Measure and record the overall efficiency (mechanical output/hydraulic input) at the maximum speed in one direction as installed in the plant at an oil temperature of 150° F. plus or minus 5° F. and at a maximum of six pressure points at equal increments starting with 750 and continuing through 2,000 psi. Repeat the above for rotation in the opposite direction.
(c) Lock the motor shaft and with one port open to reservoir, pressurize the other port to 3,000 psi for 30 seconds. Check for leakage at the assembled surfaces and the shaft seals. Repeat, pressurizing the opposite port.
(d) Test the motor for torque output in both directions at maximum rated rpm with the pressure limited to not less than 2,000 psi.
(e) With the hydraulic motor in the test stand and the maximum rated load applied to the motor shaft, increase the oil pressure to the motor at a maximum rate of 1 psi per second. Note and record the oil pressure at the instant at which the motor shaft overcomes the rated load. Repeat the above for rotation in the opposite direction.

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(f) Install the hydraulic motor in the system for running the endurance test cycle as follows: 30 minutes clockwise and 30 minutes counterclockwise direction for a total of 500 hours. Motor shall be endurance tested at maximum installed rpm, maximum rated load, and with the pressure limited to not less than 2,000 psi. For the first 50 hours, the oil temperature shall be 220° F. plus or minus 5° F. The oil temperature at the motor inlet shall be 150° F. plus or minus 5° F. for the remaining 450 hours of the test. Examine for aeration at the motor inlet ports at least every 24 hours of operation.

(g) Repeat 2(b), 2(c), 2(d), and 2(e) above.

(h) After completing the above tests, disassemble and examine the motor.

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TEST METHOD NO. 5

HYDRAULIC CYLINDER COMPONENT TEST

1. TEST APPARATUS:

(a) Test bench that includes a means of automatically cyling each cylinder.
(b) 2,000 psi hydraulic power supply (same flow as pump installed on preproduction model).
(c) Pressure source, 4,000 psi minimum.
(d) Pressure gages, temperature gages, cycle counter, recorder.
(e) A load sufficient to develop a hydraulic pressure of 1,975 psi plus or minus 25 psi.
(f) A relief valve that will limit the full flow pressure to a maximum of 2,000 psi, and will reseat at a minimum pressure of 1,850 psi.
(g) Oil conforming to MIL-L-2104, grade 10.
(h) Filter, 10 micron nominal rating, full flow, with replaceable element.

2. TEST PROCEDURE:

(a) Proof pressure.

(1) Position and hold the piston midway in the cylinder.
(2) With the opposite port vented to atmosphere, apply an oil pressure of 4,000 psi to one side of the piston for 30 seconds. Check for external leakage.
(3) Repeat 2(a) (2) to the other side of the piston.

(b) Piston leakage.

(1) Fill the cylinder with oil and cap the head end with a pressure pickup connected to the pressure recorder. Vent the rod end of the cylinder.
(2) Mechanically load the head end of the piston to not less than 2,000 psi and hold this pressure for 15 minutes at an oil temperature of 150° plus or minus 5° F.
(3) Measure and record the travel of the piston during the 15 minutes.
(4) Repeat 2(b) (1), 2(b) (2), and 2(b) (3) for the rod end of cylinder.

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(c) Packing drag.

(1) Fill the cylinder with oil at a temperature of 150° plus or minus 5° F.
(2) Open the head end port of the piston to the atmosphere with the rod under no load except normal seal and wiper drag.
(3) Pressurize the rod end of the cylinder and record the maximum pressure reached before the cylinder rod moves.
(4) Repeat the above procedure by pressurizing the head end of the cylinder.

(d) Endurance.

(1) Mount and support the cylinder on the test bench in a horizontal position or in its normal operating position. When two cylinders are used for a function in the system, and only one is being tested, the flow shall be reduced to 50 percent of that in 1(b).
(2) Mechanically connect each cylinder to a load which will develop a pressure of 1,975 psi plus or minus 25 psi on each side of the piston. The oil temperature shall be 150° plus or minus 5° F. throughout the test. The minimum rate of pressure rise developed by the cycle valve shall be 20,000 psi per second. There shall be essentially no time delay at the end of each stroke, except when adjusting the stroke control mechanism.
(3) Piston leakage. After completing endurance testing as specified in 2(d) (2), repeat the piston leakage test specified in 2(b).

(e) Disassembly and examination. Disassemble the cylinder and examine for contamination, evidence of galling, scuffing, and worn piston bearing ring permitting contact of the piston OD and cylinder ID.

(f) Subject all cylinder rods to the test specified in ASTM B117 for 30 hours. Examine the rods for corrosion.

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TEST METHOD NO. 6

HYDRAULIC SUCTION LINE TEST

1. TEST APPARATUS:

   (a) Test bench including a power supply to produce the oil pressures required.
   (b) Pressure, temperature, and cycle counter recorders.
   (c) Oil conforming to MIL-L-2104, grade 10.
   (d) Reservoir-to-pump suction line including tube, hose, clamps, and fittings as specified herein.

2. TEST PROCEDURE:

   (a) Soak the reservoir-to-pump suction line assembly in oil for 5 hours at 190° plus or minus 5° F.
   (b) Remove the line assembly from the oil bath, attach to the pressure connection of the test bench, and blank off one end. Measure and record the length of the hose. Pressurize the line assembly to 50 psi, hold this pressure for 15 seconds; then measure and record the length of the hose and examine for leakage.
   (c) Cycle the pressure in the line assembly for a total of 1,000 cycles with each cycle consisting of 7 inches of mercury vacuum for 2 seconds and 25 psia for 2 seconds. Maintain the oil and hose temperature at 150° plus or minus 5° F.
   (d) Disassemble the line assembly and examine for defects and deterioration.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-52600(ME) - Central Mixing Plant, Concrete, Mobile, 250 Cubic Yards Per Hour Capacity

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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